Semiclassical Gravitation of Prefractal Casimir Cavities: Macroscopic Effective Traces, Vernier Gravitational Radiation, Energy-Condition Switching, and Vacuum Frame Dragging

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Between infinite, flat, ideally conducting parallel plates in $(3+1)$-dimensional Minkowski spacetime, the renormalized electromagnetic Casimir stress-energy tensor $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ is conformally traceless ($T^{\mu}{}_{\mu}=0$) and invariant under in-plane Lorentz boosts ($\rho_{\rm vac}c^2+P_{\parallel}=0$). Coupled to semiclassical Einstein gravity, $G_{\mu\nu}=(8\pi G/c^4)\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$, the canonical planar vacuum therefore sources zero local Ricci scalar curvature ($R=0$), decouples from scalar-tensor breathing modes in bulk vacuum, and carries zero non-dissipative angular momentum under rigid in-plane rotation. Here we formulate the active semiclassical gravitational backreaction of an idealized zero-temperature *Canonical Prefractal Sector Cavity* (CPSC)—an azimuthally sectored, radially self-similar boundary hierarchy with short-distance length cutoff $\ell_*=Lb^{-N}$ that breaks continuous scale invariance into discrete scale invariance (scale ratio b) while breaking planar boost symmetry and modulating boundary conditions between open-aperture saddle-point fringing and parallel-sector confinement. Strictly distinguishing the cavity-integrated macroscopic effective trace $T^{\mu}{}_{\mu,\rm eff}$ from local curved-spacetime conformal trace anomalies, we combine cyclic-Bloch three-dimensional boundary-element scattering and four-pass Euclidean correlator reconstructions across conformally coupled scalar ($\xi=1/6$), minimally coupled scalar ($\xi=0$), and perfect-conductor vector electromagnetic fields with spectral linearized Einstein, Starobinsky $f(R)=R+\alpha R^2$ ($\alpha>0$), and gravito-electromagnetic solvers in Planck-normalized dimensionless units ($\epsilon_P=\ell_P/L$). Enforcing exact covariant conservation $\nabla_\mu(T^{\mu\nu}_{\rm vac}+T^{\mu\nu}_{\rm con})=0$, we establish five gravitational phenomena: (i) symmetry-filtered spin-2 gravitational radiation from a $(6,4)$ Vernier cavity whose rigid C_6 and C_4 boundaries possess zero classical mass quadrupole ($\mathcal{I}^{\rm mass}_{ij}=2.72\times10^{-16}$), yielding dimensionless vacuum-sourced strain $\tilde{h}_+=0.1145$ ($0.3857\times$ relative to an equal-area periodic grating), alongside monopole ($l=0$) scalaron radiation ($\tilde{\varphi}_0=27.122$, Yukawa-suppressed by 9.80×10^{-4} above the Starobinsky mass threshold); (ii) an axial-force-free curvature crossover at $\theta_0=25.3298^\circ$ ($|\tilde{F}_z|=0.0504$) where the scale-locked coarse-grained Ricci curvature remains finite ($\tilde{R}_{\rm eff}=-2317.8935$) while asymptotic Tolman–Komar mass respects the Weak Equivalence Principle; (iii) a discrete-scale-invariant, log-periodic Newtonian potential and tidal curvature comb with complex scaling exponents $2+i(2\pi k/\ln b)$ ($\hat{\omega}_1=5.7096$, $\hat{b}=3.0055$); (iv) reversible rotary switching of local Null and Strong Energy Conditions ($\widetilde{\rm NEC}_z\in[-579.895,6523.7421]$) and Raychaudhuri geodesic focusing strictly within Ford–Roman and Fewster worldline quantum energy inequality bounds (saturation ratio 0.1425); and (v) Cantor-stratified gravitomagnetic Lense–Thirring frame dragging ($\max|\tilde{\Omega}^{\rm vac}_{{\rm LT},z}|/\tilde{\Omega}=22931.2037$) sourced by broken vacuum boost invariance. Systematic prefractal generation ($N=1$–4) and mesh-refinement analyses confirm numerical convergence within 0.0533.

I. INTRODUCTION AND LITERATURE REVIEW

The coupling between quantum vacuum fluctuations and the classical geometry of spacetime lies at the heart of semiclassical gravity [1, 2],

$$G_{\mu\nu}\equiv R_{\mu\nu}-\frac{1}{2}g_{\mu\nu}R=\frac{8\pi G}{c^4}\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}, \tag{1}$$

where $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ denotes the renormalized expectation value of the quantum stress-energy tensor in a specified boundary and metric state. Throughout this work we retain physical constants $(\hbar,c,G)$ explicitly in dimensional equations before mapping systematically to dimensionless Planck–Casimir variables, and adopt the metric signature $(-,+,+,+)$ [2, 3]. Since Casimir's discovery that conducting boundaries modify electromagnetic zero-point modes [4, 5], the Casimir apparatus has served as the canonical arena in which renormalized quantum stresses are finite, calculable from first principles, and non-classical. Yet most results on how Casimir vacua gravitate derive from maximally symmetric planar plates separated by a single length scale a, or treat the cavity as a passive test probe inside an externally prescribed curved metric.

A. Conformal Tracelessness, Trace Anomalies vs. Effective Traces, and Planar Boost Symmetry

In their foundational image-method calculation, Brown and Maclay [6] proved that for the source-free quantized electromagnetic field between ideal parallel

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conducting plates at $z = 0$ and $z = a$ (with interaction energy per unit area $E_C/A = -\pi^2\hbar c/(720a^3)$), conformal invariance and in-plane (x, y) Poincaré symmetry uniquely fix the renormalized covariant and mixed stress-energy tensors in $(-,+,+,+)$ signature to

$$\begin{aligned}\langle\hat{T}_{\mu\nu}\rangle_{\rm flat} &= \frac{\pi^2\hbar c}{720a^4}\,{\rm diag}(-1,+1,+1,-3),\\ \langle\hat{T}^{\mu}{}_{\nu}\rangle_{\rm flat} &= {\rm diag}\big(-\rho_{\rm vac}c^2,\, P_\parallel,\, P_\parallel,\, P_\perp\big)\\ &= \frac{\pi^2\hbar c}{720a^4}\,{\rm diag}(+1,+1,+1,-3),\end{aligned}\tag{2}$$

where $T_{00} = +\rho_{\rm vac}c^2 = -\pi^2\hbar c/(720a^4) < 0$ is the negative vacuum energy density (so $T^0{}_0 = -\rho_{\rm vac}c^2 > 0$), $T_{xx} = T_{yy} = P_\parallel = -\rho_{\rm vac}c^2 > 0$ are the transverse tensions, and $T_{zz} = P_\perp = 3\rho_{\rm vac}c^2 < 0$ is the attractive normal pressure. Summing the mixed components confirms exact conformal tracelessness: $T^{\mu}{}_{\mu} = -\rho_{\rm vac}c^2 + 2P_\parallel + P_\perp = (1+1+1-3)\pi^2\hbar c/(720a^4) = 0$.

At the outset, it is essential to draw a sharp conceptual distinction between three physically distinct trace concepts in quantum field theory:

1. *Local curved-spacetime conformal (Weyl) trace anomalies*: In curved backgrounds, ultraviolet regularization breaks classical Weyl invariance and generates a local state-independent anomalous trace $\langle\hat{T}^{\mu}{}_{\mu}(x)\rangle_{\rm anom}$ proportional to geometric curvature invariants [1, 7]. As shown by Setare and Rezaeian [8] for Dirichlet plates in two-dimensional curved backgrounds, the local two-dimensional trace anomaly ($\propto R$) links boundary and background vacuum stresses directly; in $(3+1)$ dimensions, however, the local Weyl anomaly is quadratic in Riemann/Weyl curvature [7] and therefore vanishes identically ($\mathcal{O}(\epsilon_P^4)$) in flat Minkowski interior vacuum $\mathbb{R}^3 \setminus \partial\Omega$.

2. *Non-conformal local operator traces*: For scalar fields with non-conformal curvature coupling $\xi \neq 1/6$ [9–11], bulk improvement terms $\propto (1/6 - \xi)\Box\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren}$ produce a genuine local interior trace even in flat space whenever $\langle\hat{\phi}^2(\mathbf{x})\rangle_{\rm ren}$ is spatially inhomogeneous.

3. *Coarse-grained macroscopic effective trace*: Recently, Gordon [12] introduced an *effective trace framework* for self-similar Casimir hierarchies, proving that when a cavity possesses an internal geometric length cutoff ℓ_* such that the dimensionless Casimir coefficient $\mathcal{C}(\ln(d/\ell_*))$ runs logarithmically with plate separation d, the cavity-averaged macroscopic stress tensor acquires a non-vanishing *integrated effective trace* $T^{\mu}{}_{\mu,\rm eff} = -(\hbar c/d^4)\partial_{\ln d}\mathcal{C}$. Crucially, as emphasized in Ref. [12], $T^{\mu}{}_{\mu,\rm eff}$ is a coarse-grained thermodynamic / boundary-moment quantity over the cavity volume—not a local quantum trace anomaly violating $\eta^{\mu\nu}\langle\hat{T}_{\mu\nu}\rangle_{\rm EM} = 0$ at interior vacuum points.

Within linearized semiclassical gravity around flat spacetime, Eq. (2) imposes three structural degeneracies on canonical planar Casimir gravitation:

1. *Vanishing Ricci scalar and scalaron source*: Contracting Eq. (1) yields $R = -(8\pi G/c^4)\langle\hat{T}^{\mu}{}_{\mu}\rangle_{\rm ren} = 0$ across the planar gap, and constant $\mathcal{C} = -\pi^2/720$ gives $T^{\mu}{}_{\mu,\rm eff} = 0$. In quadratic $f(R) = R + \alpha R^2$ or scalar-tensor gravity [13–17], the extra scalar degree of freedom (scalaron) couples to the stress-energy trace and therefore decouples from planar conformal vacuum fluctuations.

2. *Algebraic locking of force, energy, and energy-condition violation*: Because $P_\perp = 3\rho_{\rm vac}c^2 < 0$, the planar force cannot vanish at finite separation without the vacuum energy also vanishing. Along the plate normal $k^\mu = (1,0,0,1)$, the Null Energy Condition (NEC) projection $T_{\mu\nu}k^\mu k^\nu = \rho_{\rm vac}c^2 + P_\perp = 4\rho_{\rm vac}c^2 < 0$ and Strong Energy Condition (SEC) combination $(T_{\mu\nu} - \frac{1}{2}g_{\mu\nu}T)u^\mu u^\nu = \frac{1}{2}(\rho_{\rm vac}c^2 + \sum_i P_i) = \rho_{\rm vac}c^2 < 0$ are unconditionally negative, forcing purely defocusing Raychaudhuri geodesic perturbations [18, 19].

3. *In-plane Lorentz boost invariance and rotational silence*: Parallel to the plates, the vacuum enthalpy density vanishes identically, $\rho_{\rm vac}c^2 + P_\parallel = 0$. Under rigid rotation of axisymmetric parallel disks relative to asymptotic inertial observers, the planar Casimir state acquires zero first-order laboratory momentum density $T^{0\phi}_{\rm vac} \propto v_\phi(\rho_{\rm vac}c^2 + P_\parallel) = 0$ (leaving only higher-order radiative vacuum friction $\propto \Omega^3$ on polarizable media [20, 21]), and therefore sources zero stationary gravitomagnetic Lense–Thirring field [22–25].

B. Prior Literature on Gravitational Casimir Physics, Dynamic Backgrounds, and Energy Inequalities

Extensive prior research has examined Casimir cavities placed inside externally prescribed gravitational fields. Early Fermi- and Rindler-coordinate analyses [26, 27] revealed apparent orientation-dependent forces when integrating $\langle\hat{T}_{\mu\nu}\rangle_{\rm ren}$ without mechanical support stresses. Fulling, Milton, Parashar, Romeo, Shajesh, and Wagner [28, 29] resolved this via variational principles, proving that covariant conservation $\nabla_\mu(T^{\mu\nu}_{\rm vac} + T^{\mu\nu}_{\rm con}) = 0$ causes spatial pressures to cancel in global mass integrals by the relativistic virial (Laue) theorem, yielding passive and active gravitational mass E_C/c^2 in compliance with the Weak Equivalence Principle (WEP). Complementing the *Archimedes* vacuum-weighing experiment [30], Sorge [31] applied the quasi-local Tolman and Komar mass formalisms to prove independently that a static Casimir cavity carries gravitational mass equal to E_C/c^2 and experiences a consistent Archimedes-type vacuum

buoyancy force in weak gravity, while Sorge and Wilson [32, 33] quantified tidal and nonequilibrium corrections during free fall.

More recently, time-dependent and stationary rotating background metrics have been shown to alter intracavity quantum states significantly. Sorge [34] demonstrated a *gravitational memory of the Casimir effect*, wherein a transient anisotropic spacetime pulse leaves a permanent shift in cavity vacuum energy. De Oliveira, Moreira, and Céleri [35] derived particle creation and parametric-amplification resonances in the dynamical Casimir effect (DCE) when an oscillating cavity interacts with a passing gravitational wave. In stationary curved geometries, Zhang [36] proved that background frame dragging (off-diagonal metric components g_{0i} as in Kerr spacetime) renormalizes Casimir energy and thermal free energy, while Pourhassan [17] and Garattini and Tzikas [37] analyzed rotating Casimir wormholes and perturbative scalaron stability in $f(R) = R + \alpha R^2$ gravity. Crucially, Refs. [34–36] address the *passive test-field problem*—how an imposed external GW or Kerr metric modifies the quantum field inside a simple cavity—whereas Refs. [17, 37] study self-consistent exotic wormhole throats. Conversely, any interpretation of negative Casimir energy as "exotic matter" is strictly constrained by Quantum Energy Inequalities (QEIs): Ford and Roman [38, 39] and Fewster and Eveson [40, 41] proved that along any smooth timelike worldline sampled over timescale τ, negative energy densities satisfy $\int_{-\infty}^{\infty}\langle\hat{T}_{00}(t)\rangle|f(t)|^2\mathrm{d}t \geq -C/\tau^4$, preventing macroscopic causal pathologies while requiring positive compensating stress concentrations near boundaries [19, 42].

C. Fractal Casimir Precedents, Aperture Stress Inversion, and Novel Contributions

In flat Minkowski spacetime without gravity, studies of quantum fields on fractals [43] and self-similar plate configurations by Shajesh *et al.* [44, 45] established explicit recursion methods for Cantor-type plate stacks and Sierpinski geometries, uncovering alternating-sign finite-generation corrections and convergence plateaus as prefractal depth $N \to \infty$, while Gordon [12] established the one-dimensional thermodynamic effective trace identity under discrete scale invariance (DSI) [46]—a symmetry celebrated in general relativity since Choptuik's critical gravitational collapse [47] and Calcagni's multifractal gravity [48]. Independently, whereas mirror-symmetric bodies in vacuum always attract [49], Levin *et al.* [50] proved via exact boundary-element scattering [51–53] that an elongated conductor aligned with a metallic slot aperture in vacuum ($\varepsilon = 1$) experiences repulsive saddle-point axial Casimir force ($F_z > 0$).

Unlike Refs. [12, 44, 45] (which evaluate static Casimir energies or 1D effective traces without solving gravitational field equations) and unlike Refs. [31, 34–36] (which place planar cavities in external metrics), the present work solves the *active three-dimensional semiclassical backreaction problem* for a rotating, azimuthally sectored prefractal cavity in flat background spacetime. Specifically, we introduce the zero-temperature *Canonical Prefractal Sector Cavity* (CPSC) and establish the following five new gravitational results:

1. **Active Vernier Spin-2 Radiation and Massive $f(R)$ Scalaron Dispersion (Sec. IV)**: Whereas de Oliveira *et al.* [35] and Sorge [34] study cavity excitation by an incoming GW, we prove from cyclic scattering group theory (Appendix B) that steady adiabatic rotation ($\Omega \ll \pi c/d_0$, zero photon creation) of a rigid C_6 rotor over a C_4 prefractal stator eliminates all classical rigid-body $l = 2$ mass-quadrupole radiation while generating a rotating $m = 6 - 4 = 2$ vacuum energy wave that actively radiates spin-2 GWs at 6Ω, alongside $l = 0$ Starobinsky scalaron waves ($\alpha > 0$) governed by Klein–Gordon Yukawa screening.

2. **Axial-Force-Free Effective Curvature vs. Global Tolman–Komar WEP Recovery (Sec. V)**: Extending Sorge's Tolman/Komar analysis [31] and Fulling *et al.* [28] to repulsive-to-attractive aperture crossovers [50], we prove that at the zero-axial-force angle θ_0 where vertical wall loads vanish ($F_z(\theta_0) = 0$), logarithmic scale locking ($\partial_{\ln|z|}\mathcal{C} = 3\mathcal{C}$) preserves a finite interior coarse-grained curvature $R_{\mathrm{eff}}(\theta_0) \neq 0$ while global Komar mass strictly equals $E_C(\theta)/c^2$.

3. **Discrete-Scale-Invariant Log-Periodic Newtonian Metric and Tidal Comb (Sec. VI)**: Extending flat-space Cantor hierarchies [12, 44] to the 3D linearized Einstein Poisson equation, we demonstrate a switchable log-periodic gravitational potential h_{00} and tidal Riemann comb R_{0z0z} with complex scaling dimensions $2 + i(2\pi k/\ln b)$.

4. **Rotary NEC/SEC Geodesic Switching Bounded by Worldline QEIs (Sec. VII)**: We show that mechanical rotation reversibly switches axial slot channels between Raychaudhuri defocusing ($\mathrm{NEC}_z < 0$) and converging microlensing ($\mathrm{NEC}_z > 0, \mathrm{SEC} > 0$) while strictly obeying Ford–Roman and Fewster worldline QEIs [39, 41].

5. **Active Vacuum Lense–Thirring Vorticity from Broken Boost Invariance (Sec. VIII)**: Contrasting with external-metric frame dragging [36] and framed within Costa and Natário's gauge-invariant gravito-electromagnetic hierarchy [24, 25], we prove that azimuthal slot walls break vacuum boost symmetry ($\rho_{\mathrm{vac}}c^2 + P_\phi \neq 0$), endowing the rotating CPSC with genuine laboratory-frame vacuum momentum $T^{0\phi}_{\mathrm{vac}}$ and Cantor-stratified Lense–Thirring precession $\boldsymbol{\Omega}^{\mathrm{vac}}_{\mathrm{LT}}$.

The remainder of this article is organized as follows. Section II defines Planck–Casimir units, the CPSC ge-

ometry, the exact 1D/flat reduction of the 3D macroscopic effective trace, and covariant conservation. Section III details our boundary-element, four-pass voxel correlator, and spectral PDE solvers, together with prefractal ($N = 1$–4) and mesh-refinement convergence tests (Sec. III D). Sections IV–VIII present derivations and numerical benchmarks for the five effects. Section IX examines covariant consistency and physical scale estimates, and Sec. X summarizes our conclusions and caveats.

II. CANONICAL PREFRACTAL SECTOR CAVITY AND COVARIANT EFFECTIVE TRACE FORMALISM

A. Planck–Casimir Dimensionless Units

Let L denote the outer radius of the cavity (carrying physical dimension of length). Zero-temperature Casimir stresses scale with $\hbar c/L^4$ while Einstein's coupling introduces G/c^4; their product defines the squared Planck length $\ell_P^2 = \hbar G/c^3 \approx 2.612 \times 10^{-70}\ \mathrm{m}^2$ and dimensionless Planck–cavity ratio $\epsilon_P \equiv \ell_P/L$. To prevent floating-point underflow while preserving exact physical scaling, we define dimensionless tilde quantities:

$$\tilde{x}^\mu = \frac{x^\mu}{L}, \qquad \tilde{\kappa} = \kappa L, \qquad \tilde{E}_C = \frac{L}{\hbar c} E_C, \qquad \tilde{F}_z = \frac{L^2}{\hbar c} F_z,$$
$$\tilde{T}_{\mu\nu} = \frac{L^4}{\hbar c}\langle \hat{T}_{\mu\nu}\rangle_{\mathrm{ren}}, \qquad \tilde{R}_{\mu\nu} = \frac{L^4}{\ell_P^2} R_{\mu\nu}, \qquad \tilde{\bar{h}}_{\mu\nu} = \frac{L^2}{\ell_P^2}\bar{h}_{\mu\nu}, \tag{3}$$

where $\bar{h}_{\mu\nu} \equiv h_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\eta^{\alpha\beta}h_{\alpha\beta}$ is the trace-reversed metric perturbation in Lorenz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$. In these units, the linearized semiclassical Einstein equation takes the parameter-free form

$$\tilde{\Box}\tilde{\bar{h}}_{\mu\nu}(\tilde{x}) = \left(-\frac{\partial^2}{\partial \tilde{x}_0^2} + \tilde{\nabla}^2\right)\tilde{\bar{h}}_{\mu\nu}(\tilde{x}) = -16\pi\, \tilde{T}^{\mathrm{tot}}_{\mu\nu}(\tilde{x}), \tag{4}$$

with local dimensionless Ricci scalar $\tilde{R}(\tilde{x}) = -8\pi \tilde{T}^{\mu}{}_{\mu,\mathrm{tot}}(\tilde{x})$. Physical SI values at any reference length scale L follow via $h_{\mu\nu} = \epsilon_P^2 \tilde{h}_{\mu\nu}$ and $R = (\epsilon_P^2/L^2)\tilde{R}$.

B. Geometry of the Canonical Prefractal Sector Cavity

We consider two disjoint, rigid, ideally lossless boundary assemblies in $(3+1)$-dimensional Minkowski vacuum at $T = 0$ K spanning the cylindrical annulus $r \in [r_{\mathrm{in}}, r_{\mathrm{out}}] = [0.25L, 1.00L]$ ($\tilde{r} \in [0.25, 1.00]$, radial span $\Delta\tilde{r} = 0.75$; Fig. 1):

1. *Recursive radial Cantor hierarchy $\mathcal{C}_{N,b}$ and cutoff length ℓ_**: Given prefractal generation order $N \in \{0, 1, 2, 3, 4\}$ and scaling factor $b > 2$ (retention ratio $\eta = 1/b$ and central removal fraction $\gamma = 1 - 2/b$, yielding the middle-third Cantor set at $b = 3$), generation $m \in \{1, \ldots, N\}$ defines 2^{m-1} concentric aperture slots ($N_{\mathrm{tracks}} = 2^N - 1$ total slots for $N \geq 1$) of dimensionless radial width

$$\tilde{W}_m = \Delta\tilde{r}\left(1 - \frac{2}{b}\right) b^{-(m-1)}, \qquad m = 1, \ldots, N, \tag{5}$$

terminating at the physical short-distance prefractal cutoff length $\ell_* \equiv \Delta r\, b^{-N}$ (dimensionless $\tilde{\ell}_* = \ell_*/L = \Delta\tilde{r}\, b^{-N}$). Centered inside each slot $\tilde{r} \in [\tilde{r}_{c,k} - \tilde{W}_m/2, \tilde{r}_{c,k} + \tilde{W}_m/2]$ is an upper vane of width $\tilde{w}_m = \chi\tilde{W}_m$ ($\chi = 0.50$), yielding self-similar radial clearances $\tilde{g}_{r,m} = \frac{1}{2}(1-\chi)\tilde{W}_m = \tilde{g}_{r,1} b^{-(m-1)}$.

2. *Coplanar versus homothetic (r, z) vertical staging*: In the *coplanar* CPSC, lower aperture rims occupy $\tilde{z} \in [-\tilde{\tau}_0, 0]$ ($\tilde{\tau}_0 = 0.03$), upper vanes span $\tilde{z} \in [\tilde{z}_{\mathrm{vane}}, \tilde{z}_{\mathrm{vane}} + \tilde{h}_0]$ ($\tilde{h}_0 = 0.12$, baseline $\tilde{z}_{\mathrm{vane}} = -0.10$), and lower facing sectors sit at $\tilde{z}_{\mathrm{top}} = -0.12$ (vertical facing separation $\tilde{d}_0 = d_0/L = 0.02$). In the *homothetic (r, z)-staged* CPSC, cross-sectional dimensions and vertical aperture centers scale geometrically as

$$\tilde{z}_{\mathrm{stage},m} = -\tilde{Z}_1 b^{-(m-1)}, \qquad \tilde{h}_{\mathrm{rim},m} = \tilde{\tau}_1 b^{-(m-1)}, \tag{6}$$

with $\tilde{Z}_1 = 0.27$ and $\tilde{\tau}_1 = 0.054$, so rescaling $|\tilde{z}| \to |\tilde{z}|/b$ maps generation m onto generation $m+1$.

3. *Azimuthal multipole constellations $(M_{\mathrm{up}}, M_{\mathrm{low}})$ and controls*: Let M_{up} and M_{low} denote the discrete cyclic symmetry orders of the upper rotor and lower stator. Rotation by θ transitions vanes between open-aperture alignment ($\theta_{\mathrm{open}} = 0$, saddle-point fringing [50]) and facing-sector alignment ($\theta_{\mathrm{face}} = \pi/M_{\mathrm{low}}$, parallel-plate confinement [4, 6]). We analyze three constellations: (A) $(M_{\mathrm{up}}, M_{\mathrm{low}}) = (4, 4)$ breathing symmetry; (B) $(M_{\mathrm{up}}, M_{\mathrm{low}}) = (6, 4)$ Vernier quadrupole symmetry; and (C) chiral skew-sector $(4, 4)$ symmetry with azimuthal bevel $\delta\phi_m$. Each prefractal configuration is benchmarked against two iso-measure controls on $\tilde{r} \in [0.25, 1.00]$: a flat parallel annulus (`Ctrl_FlatAnnulus`) and an equal-slot-area single-scale periodic grating (`Ctrl_PeriodicGrating`) with N_{tracks} uniform slots of width $\tilde{W}_{\mathrm{unif}} = N_{\mathrm{tracks}}^{-1}\sum_k \tilde{W}_k$.

C. Quantum Field Models, Exact Reduction of the Macroscopic Effective Trace, and Covariant Conservation

To separate spin and local conformal properties, we evaluate three massless $T = 0$ K quantum field models on the CPSC boundaries $\partial\Omega = \partial\Omega_{\mathrm{up}} \cup \partial\Omega_{\mathrm{low}}$:

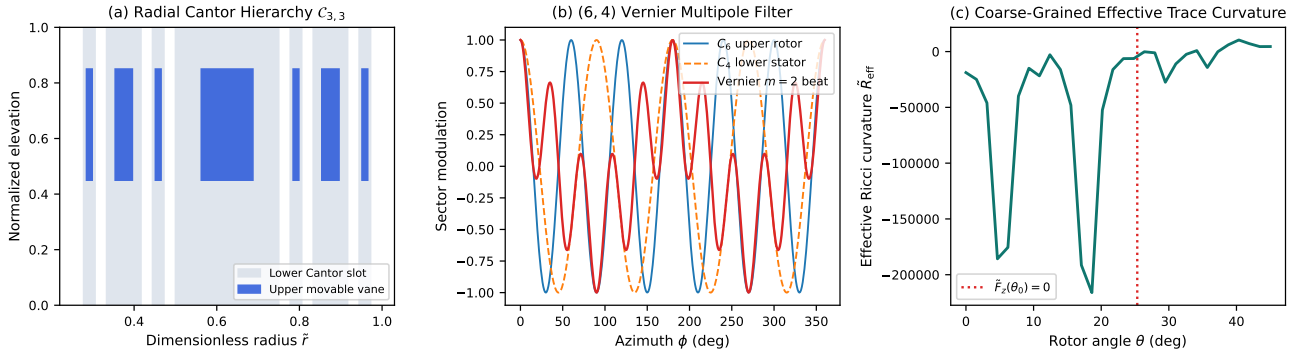


FIG. 1. Zero-temperature geometry and macroscopic effective-trace mechanism of the Canonical Prefractal Sector Cavity (CPSC) in dimensionless coordinates $\tilde{x}^\mu = x^\mu/L$ ($\epsilon_P = \ell_P/L$). (a) Analytic cross-sectional profile of the recursive middle-third ($b = 3, N = 3$) radial Cantor slot hierarchy $\mathcal{C}_{3,3}$ spanning $\tilde{r} \in [0.25, 1.00]$, showing lower stator aperture slots (gray) and centered upper rotor vanes (blue, fill ratio $\chi = 0.50$). (b) Analytic azimuthal multipole indicator modulation of the $(M_{\text{up}}, M_{\text{low}}) = (6, 4)$ Vernier constellation, illustrating how the product of C_6 rotor (solid blue) and C_4 stator (dashed orange) profiles synthesizes a pure $m = 2$ beat wave (red). (c) First-principles boundary-element numerical sweep of the dimensionless coarse-grained Ricci curvature $\tilde{R}_{\text{eff}}(\theta) = -8\pi\tilde{T}^{\mu}{}_{\mu,\text{eff}}(\theta)$ for ideal PEC electromagnetism across rotor angle $\theta \in [0°, 45°]$ in Constellation A $(4, 4)$, highlighting the axial-force-free crossover $\tilde{F}_z(\theta_0) = 0$ (vertical dotted line).

1. *Conformal massless real scalar ($\xi = 1/6$)*: Governed by the Callan–Coleman–Jackiw improved stress-energy tensor [1, 9],

$$\hat{T}^{(\xi)}_{\mu\nu} = \partial_\mu\hat{\phi}\,\partial_\nu\hat{\phi} - \frac{1}{2}\eta_{\mu\nu}(\partial\hat{\phi})^2 + \xi\big(\eta_{\mu\nu}\Box - \partial_\mu\partial_\nu\big)\hat{\phi}^2, \quad (7)$$

satisfying $\eta^{\mu\nu}\langle\hat{T}^{(1/6)}_{\mu\nu}(\mathbf{x})\rangle_{\text{ren}} = 0$ identically on shell ($\Box\hat{\phi} = 0$) everywhere in interior vacuum $\mathbb{R}^3 \setminus \partial\Omega$.

2. *Minimally coupled massless real scalar ($\xi = 0$)*: Carries an intrinsic local bulk trace $\eta^{\mu\nu}\langle\hat{T}^{(0)}_{\mu\nu}(\mathbf{x})\rangle_{\text{ren}} = -\frac{1}{2}\Box\langle\hat{\phi}^2(\mathbf{x})\rangle_{\text{ren}} \neq 0$ whenever $\langle\hat{\phi}^2(\mathbf{x})\rangle_{\text{ren}}$ varies spatially [10, 11].

3. *Ideal 0 K PEC vector electromagnetism*: Governed by the renormalized Maxwell stress-energy tensor

$$\langle\hat{T}_{00}\rangle_{\text{EM}} = \frac{1}{2}\left\langle\varepsilon_0\mathbf{E}^2 + \mu_0\mathbf{H}^2\right\rangle_{\text{ren}},$$
$$\langle\hat{T}_{ij}\rangle_{\text{EM}} = \frac{1}{2}\delta_{ij}\left\langle\varepsilon_0\mathbf{E}^2 + \mu_0\mathbf{H}^2\right\rangle_{\text{ren}} - \left\langle\varepsilon_0 E_iE_j + \mu_0 H_iH_j\right\rangle_{\text{ren}}, \quad (8)$$

which is pointwise traceless ($\eta^{\mu\nu}\langle\hat{T}_{\mu\nu}(\mathbf{x})\rangle_{\text{EM}} = 0$) at every interior point $\mathbf{x} \in \mathbb{R}^3 \setminus \partial\Omega$ (verified numerically in our CPSC solver to relative residual 2.49×10^{-16}).

To derive the three-dimensional macroscopic effective trace and verify its exact dimensional and planar limits [12], express the dimensional zero-temperature Casimir interaction energy across vertical separation $d = |z|$ and outer radius R (annular area $A_{\text{ann}} = \pi(r^2_{\text{out}} - r^2_{\text{in}}) \propto R^2$) at fixed physical prefractal cutoff length $\ell_* = \Delta r\, b^{-N}$ as

$$\frac{E_C(\theta, d, R; \ell_*)}{A_{\text{ann}}} = \frac{\hbar c}{d^3}\,\mathcal{C}\left(\theta, \ln\frac{d}{\ell_*}, \frac{R}{\ell_*}\right), \quad (9)$$

where every argument of $\mathcal{C}$ is manifestly dimensionless. Defining the coarse-grained vacuum energy density $\rho_{\text{vac}}c^2 \equiv E_C/V_{\text{cav}}$ and virtual-work macroscopic cavity pressures over reference volume $V_{\text{cav}} = A_{\text{ann}}d$ at fixed ℓ_* yields

$$\rho_{\text{vac}}c^2 = \frac{\hbar c}{d^4}\mathcal{C}, \qquad P_z = -\frac{1}{A_{\text{ann}}}\frac{\partial E_C}{\partial d}\bigg|_{R,\ell_*} = \frac{\hbar c}{d^4}\left(3\mathcal{C} - \frac{\partial\mathcal{C}}{\partial\ln d}\right),$$
$$P_r + P_\phi = -\frac{1}{d}\frac{R}{A_{\text{ann}}}\frac{\partial E_C}{\partial R}\bigg|_{d,\ell_*} = -\frac{\hbar c}{d^4}\left(2\mathcal{C} + \frac{\partial\mathcal{C}}{\partial\ln R}\right). \quad (10)$$

Summing the mixed coarse-grained components $T^{\mu}{}_{\mu,\text{eff}} \equiv -\rho_{\text{vac}}c^2 + P_r + P_\phi + P_z$ cancels the $3\mathcal{C} - \mathcal{C} - 2\mathcal{C} = 0$ algebraic terms identically, leaving in dimensional and dimensionless forms (with $\tilde{d} = |\tilde{z}|$):

$$T^{\mu}{}_{\mu,\text{eff}} = -\frac{\hbar c}{d^4}\left(\frac{\partial\mathcal{C}}{\partial\ln d} + \frac{\partial\mathcal{C}}{\partial\ln R}\right)\bigg|_{\ell_*},$$
$$\tilde{T}^{\mu}{}_{\mu,\text{eff}}(\theta, \tilde{z}) = -\frac{1}{\tilde{V}_{\text{cav}}}\mathcal{D}_{\text{cav}}\tilde{E}_C = -\frac{1}{|\tilde{z}|^4}\left(\frac{\partial\mathcal{C}}{\partial\ln|\tilde{z}|} + \frac{\partial\mathcal{C}}{\partial\ln\tilde{R}}\right)\bigg|_{\tilde{\ell}_*}, \quad (11)$$

where $\mathcal{D}_{\text{cav}} \equiv 1 + |\tilde{z}|\partial_{|\tilde{z}|}|_{\tilde{\ell}_*} + \tilde{R}\partial_{\tilde{R}}|_{\tilde{\ell}_*}$. Equation (11) satisfies three exact consistency checks: (i) in the laterally translation-invariant 1D plate-stack limit ($\partial_{\ln R}\mathcal{C} = 0$), Eq. (11) reduces identically—including sign and prefactor—to Gordon's formula $T^{\mu}{}_{\mu,\text{eff}} = -(\hbar c/d^4)\partial_{\ln d}\mathcal{C}$ [12]; (ii) for featureless parallel plates where $\mathcal{C} = -\pi^2/720 = \text{const}$, Eq. (10) reproduces Casimir's $E_C/A = -\pi^2\hbar c/(720d^3)$ law, $P_\parallel = -\rho_{\text{vac}}c^2$, $P_\perp = 3\rho_{\text{vac}}c^2$, and $T^{\mu}{}_{\mu,\text{eff}} = 0$; and (iii) by Euler homogeneity under simultaneous dilation of all lengths including ℓ_*, $\mathcal{D}_{\text{cav}}\tilde{E}_C = -\tilde{\ell}_*\partial_{\tilde{\ell}_*}\tilde{E}_C$, demonstrating that $\tilde{T}^{\mu}{}_{\mu,\text{eff}}$

originates from boundary work against the internal pre-fractal cutoff $\tilde{\ell}_*$. Note also how this $T = 0$ K zero-point trace differs from the finite-temperature fractal thermal sector [12, 43]: for thermal radiation on a fractal of spectral dimension d_s (the effective dimension governing diffusion and Laplacian eigenvalue density $\propto \omega^{d_s-1}$), the coarse-grained thermal energy density $\rho_{\rm th} = U_{\rm th}/V_s$ per spectral volume V_s and isotropic pressure $p_{\rm th} = \rho_{\rm th}/d_s$ yield $T^{\mu}{}_{\mu,\rm th} = -\rho_{\rm th} + 3p_{\rm th} = \rho_{\rm th}(3/d_s - 1)$. Because our CPSC operates strictly at $T = 0$ K ($\rho_{\rm th} = 0$) in $(3+1)$-dimensional bulk vacuum, $T^{\mu}{}_{\mu,\rm th}$ vanishes identically and only the zero-point logarithmic derivative (11) is active. Under discrete scale invariance ($\tilde{z} \to \tilde{z}/b$), $\mathcal{C}$ expands in log-periodic harmonics:

$$\mathcal{C}\left(\theta, \ln\frac{|\tilde{z}|}{\tilde{\ell}_*}\right) = \sum_{k=-\infty}^{\infty} \mathcal{C}_k(\theta) \exp\left(i\omega_k \ln\frac{|\tilde{z}|}{\tilde{\ell}_*}\right), \qquad \omega_k = \frac{2\pi k}{\ln b}. \tag{12}$$

Finally, because the left-hand side of Eq. (4) is divergence-free in Lorenz gauge ($\partial^{\mu}\tilde{\Box}\tilde{\bar{h}}_{\mu\nu} = 0$), mathematical consistency of linearized general relativity requires exact conservation of the *total* stress-energy tensor:

$$\partial_{\mu}\tilde{T}^{\mu\nu}_{\rm tot} = \partial_{\mu}\left(\tilde{T}^{\mu\nu}_{\rm vac} + \tilde{T}^{\mu\nu}_{\rm con}\right) = 0, \tag{13}$$

where $\tilde{T}^{\mu\nu}_{\rm con}$ represents the mechanical constraint stresses of the rigid support assembly balancing Casimir forces $\tilde{F}_z(\theta)$ and torques $\tilde{\tau}_\theta(\theta) = -\partial_\theta \tilde{E}_C(\theta)$ [28, 29, 31], as well as external shaft work $\partial_{\tilde{t}}\tilde{E}_C = -\tilde{\Omega}\tilde{\tau}_\theta$ when $\tilde{\tau}_\theta \neq 0$. Appendix A proves how Eq. (13) enforces Laue's virial theorem globally while preserving local interior vacuum curvature and radiative multipoles.

III. FIRST-PRINCIPLES SCATTERING, VOXEL CORRELATOR, AND CONVERGENCE VERIFICATION

To evaluate $\tilde{E}_C$, $\tilde{T}_{\mu\nu}(\tilde{\mathbf{x}})$, $\tilde{T}^{\mu}{}_{\mu,\rm eff}$, and all induced gravitational perturbations without proximity-force approximations, we implement three coupled solvers verified by systematic convergence tests.

A. 3D Cyclic-Bloch Boundary-Element Scattering and Scale-Derivative Engine

Along the Wick-rotated imaginary wavenumber axis $\tilde{\kappa} = \xi L/c \in (0, \infty)$, the zero-temperature Casimir interaction energy between disjoint upper (U) and lower (L) boundary manifolds $\partial\Omega_{\rm U} \cup \partial\Omega_{\rm L}$ is evaluated via the exact surface-scattering log-determinant [51, 52]:

$$\tilde{E}_C(\theta, \tilde{z}) = \frac{\sigma_{\rm field}}{2\pi}\int_0^{\infty} {\rm d}\tilde{\kappa}\, \ln\det\big[\mathbf{I} - \mathbf{M}(\tilde{\kappa}; \theta, \tilde{z})\big], \tag{14}$$

where $\sigma_{\rm field} = \frac{1}{2}$ for a real scalar field and $\sigma_{\rm field} = 1$ for two-polarization electromagnetism, and $\mathbf{M}(\tilde{\kappa}) = \mathbf{\Gamma}^{-1}_{\rm UU}\mathbf{\Gamma}_{\rm UL}\mathbf{\Gamma}^{-1}_{\rm LL}\mathbf{\Gamma}_{\rm LU}$ subtracts single-body ultraviolet self-energies identically at every $\tilde{\kappa}$. Boundary surfaces are meshed into N_p quadrilateral/annular panels $\{\mathcal{P}_a\}_{a=1}^{N_p}$ with piecewise-constant basis functions; diagonal self-interaction integrals are regularized analytically by replacing $|\tilde{\mathbf{r}} - \tilde{\mathbf{r}}'|$ at coincidence with the equal-area disk radius $\tilde{\sigma}_a = \sqrt{\tilde{A}_a/(16\pi)}$:

- *Scalar Dirichlet kernel*: Single-layer Euclidean Yukawa kernel $G_0(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'; \tilde{\kappa}) = \exp(-\tilde{\kappa}|\tilde{\mathbf{r}} - \tilde{\mathbf{r}}'|)/(4\pi|\tilde{\mathbf{r}} - \tilde{\mathbf{r}}'|)$ (N_p degrees of freedom).

- *PEC vector-dyadic kernel*: Electric-field integral equation (EFIE) projecting two tangential surface-current basis vectors $(\hat{\mathbf{t}}_{a,1}, \hat{\mathbf{t}}_{a,2})$ per panel onto the Euclidean dyadic Green's tensor $\mathbf{G}(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'; \tilde{\kappa}) = \big(\mathbf{I} - \tilde{\kappa}^{-2}\tilde{\nabla}\tilde{\nabla}\big)G_0(\tilde{\mathbf{r}}, \tilde{\mathbf{r}}'; \tilde{\kappa})$ ($2N_p$ degrees of freedom).

For constellations with cyclic greatest common divisor $M_g = \gcd(M_{\rm up}, M_{\rm low}) > 1$ ($M_g = 4$ for $(4,4)$ and $M_g = 2$ for $(6,4)$), discrete Fourier projection into M_g decoupled Bloch blocks $\mathbf{M}^{(\ell)}(\tilde{\kappa}) = \sum_{q=0}^{M_g-1} e^{2\pi i\ell q/M_g}\mathbf{M}_{0,q}(\tilde{\kappa})$ accelerates LU factorization by M_g^3 without approximation (verified against full-ring dense LU to $< 10^{-11}$ relative error). Integrals over $\tilde{\kappa}$ use log-mapped Gauss–Legendre quadrature on $\tilde{\kappa} \in [10^{-2}, 1.5 \times 10^2]$, and parametric derivatives ($\tilde{F}_z$, $\tilde{\tau}_\theta$, $\mathcal{D}_{\rm cav}\tilde{E}_C$) use centered finite differences of logarithmic step $\Delta \ln d = 0.015$.

B. Four-Pass Inclusion–Exclusion 3D Euclidean Correlator Voxel Mapping

To reconstruct the spatially resolved renormalized stress tensor $\tilde{T}_{\mu\nu}(\tilde{\mathbf{x}}; \theta)$ at interior vacuum points $\tilde{\mathbf{x}} \in \Omega_{\rm grid} \subset \mathbb{R}^3 \setminus \partial\Omega$ (masked outside solid conductor volumes by safety clearance $\tilde{\delta}_{\rm wall} = 0.008$), we evaluate the scattered Euclidean Green's correlator $\Delta G(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) \equiv G^{\rm full}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) - G_0(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa})$. Subtracting free-space coincidence divergences (G_0) directly and isolating mutual inter-assembly stresses via four-pass inclusion–exclusion yields

$$\Delta\tilde{T}^{\rm mut}_{\mu\nu}(\tilde{\mathbf{x}}; \theta) = \tilde{T}^{\rm U+L}_{\mu\nu}(\tilde{\mathbf{x}}; \theta) - \tilde{T}^{\rm U}_{\mu\nu}(\tilde{\mathbf{x}}; \theta) - \tilde{T}^{\rm L}_{\mu\nu}(\tilde{\mathbf{x}}) + \tilde{T}^{\rm vac}_{\mu\nu}(\tilde{\mathbf{x}}), \tag{15}$$

alongside the boundary-renormalized total interior tensor $\tilde{T}^{\rm ren}_{\mu\nu}(\tilde{\mathbf{x}}; \theta) = \tilde{T}^{\rm U+L}_{\mu\nu}(\tilde{\mathbf{x}}; \theta) - \tilde{T}^{\rm vac}_{\mu\nu}(\tilde{\mathbf{x}})$, evaluated from the LU-factorized scattering inverse $\mathbf{\Gamma}^{-1}(\tilde{\kappa})$:

$$\Delta G_{\rm ren}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}'; \tilde{\kappa}) = -\sum_{a,b} g_0(\tilde{\mathbf{x}}, \mathcal{P}_a; \tilde{\kappa})\big[\mathbf{\Gamma}^{-1}(\tilde{\kappa})\big]_{ab} g_0(\mathcal{P}_b, \tilde{\mathbf{x}}'; \tilde{\kappa}). \tag{16}$$

Analytical spatial derivatives $\partial_i\partial'_j\Delta G_{\rm ren}|_{\tilde{\mathbf{x}}'=\tilde{\mathbf{x}}}$ and $\tilde{\kappa}^2\Delta G_{\rm ren}(\tilde{\mathbf{x}}, \tilde{\mathbf{x}}; \tilde{\kappa})$ assemble the scalar (7) and dyadic electromagnetic (8) stress tensors at each voxel.

C. 3D Spectral Linearized Einstein, Scalaron, and Gravitomagnetic PDE Solvers

From $\tilde{T}^{\rm tot}_{\mu\nu}(\tilde{\mathbf{x}},\tilde{t})$, gravitational observables are computed across three sectors:

1. *Far-zone radiative multipole integration*: In the slow-motion regime ($\tilde{\omega} = \omega L/c \ll 1$), transverse-traceless spin-2 waveforms along $\hat{\mathbf{n}} = (\sin\vartheta\cos\varphi, \sin\vartheta\sin\varphi, \cos\vartheta)$ follow the conserved quadrupole formula [2, 3],

$$\tilde{h}^{\rm TT}_{ij}(\tilde{t},\tilde{r}\hat{\mathbf{n}}) = \frac{2}{\tilde{r}}\Lambda_{ij,kl}(\hat{\mathbf{n}})\frac{\mathrm{d}^2\tilde{\mathcal{Q}}^{\rm tot}_{kl}}{\mathrm{d}\tilde{t}^2}(\tilde{t}-\tilde{r}),$$
$$\tilde{\mathcal{Q}}^{\rm tot}_{kl}(\tilde{t}) = \int \tilde{T}^{\rm tot}_{00}(\tilde{\mathbf{x}},\tilde{t})\left(\tilde{x}_k\tilde{x}_l - \frac{1}{3}\tilde{\mathbf{x}}^2\delta_{kl}\right)\mathrm{d}^3\tilde{x}, \quad (17)$$

cross-checked against $2\int(\tilde{T}^{\rm tot}_{kl})^{\rm TF}\mathrm{d}^3\tilde{x}$ via Appendix A. In linearized $f(R) = R + \alpha R^2$ gravity ($\alpha > 0$, dimensionless scalaron mass $\tilde{m}_\varphi = L/\sqrt{6\alpha}$), Fourier modes $\tilde{\mathcal{T}}_{\rm eff}(\tilde{\omega})$ of the trace source evolve via the exact retarded/Yukawa Klein–Gordon Green's function [16]:

$$\tilde{\varphi}(\tilde{t},\tilde{r}) = -\frac{2\alpha_\varphi}{\tilde{r}}\int_0^\infty \mathrm{d}\tilde{\omega}\,\mathrm{Re}\left[\tilde{\mathcal{T}}_{\rm eff}(\tilde{\omega})\,\mathcal{G}_\varphi(\tilde{\omega},\tilde{r})e^{-i\tilde{\omega}\tilde{t}}\right],$$
$$\mathcal{G}_\varphi(\tilde{\omega},\tilde{r}) = \begin{cases} e^{i\sqrt{\tilde{\omega}^2-\tilde{m}_\varphi^2}\,\tilde{r}}, & \tilde{\omega} > \tilde{m}_\varphi, \\ e^{-\sqrt{\tilde{m}_\varphi^2-\tilde{\omega}^2}\,\tilde{r}}, & \tilde{\omega} \le \tilde{m}_\varphi, \end{cases} \quad (18)$$

with canonical coupling $\alpha_\varphi = 1/\sqrt{6}$.

2. *Near-zone 3D isolated-boundary Poisson solver*: Elliptic equations $\tilde{\nabla}^2\tilde{\bar{h}}_{\mu\nu} = -16\pi\tilde{T}^{\rm tot}_{\mu\nu}$ and $\tilde{\nabla}^2\tilde{h}_{0i} = +16\pi\tilde{T}^{0i}_{\rm tot}$ with open falloff $\tilde{\bar{h}}_{\mu\nu} \to 0$ as $|\tilde{\mathbf{x}}| \to \infty$ are solved via 2× zero-padded Hockney–Eastwood 3D Fast Fourier Transform convolution with the spherical-equivalent regularized kernel $1/(4\pi|\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'|)$ (reproducing analytical uniform-ball potentials within $< 3\%$), using fourth-order central stencils for $\tilde{R}_{0z0z} = -\frac{1}{2}\partial^2_{\tilde{z}}\tilde{h}_{00}$ and $\tilde{\mathbf{B}}^{\rm vac}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0$.

3. *Raychaudhuri and QEI quadrature*: Geodesic expansion shifts and Lorentzian-weighted worldline QEI integrals are evaluated by adaptive quadrature along axial rays and stationary azimuthal observer worldlines.

D. Regularization, Truncation Error Control, and Prefractal Convergence ($N \to \infty$)

To ensure reproducibility and verify that all five effects persist under mesh refinement and increasing prefractal order N [12, 44], Fig. 2 reports four systematic convergence benchmarks:

1. *Prefractal generation plateau ($N = 1, 2, 3, 4$)*: As N increases at fixed facing gap $\tilde{d}_0 = 0.02$, generation-N Cantor slots narrow as $\tilde{W}_N \propto b^{-(N-1)}$. Once $\tilde{W}_N \lesssim \tilde{d}_0$, evanescent Yukawa modes ($\tilde{\kappa} \sim 1/\tilde{d}_0$) smooth over finer sub-slots, causing finite-generation energy increments to decay geometrically (Fig. 2a,b): the relative Cauchy difference $|\tilde{E}_C(4) - \tilde{E}_C(3)|/|\tilde{E}_C(4)|$ drops to 0.0665, while the force-free crossover angle stabilizes at $\theta_0(N=4) = 25.7528°$, within 0.423° of $\theta_0(N=3) = 25.3298°$.

2. *Wick quadrature, BEM panel, and 3D voxel truncation uncertainties*: At our production quadrature order $N_\kappa = 14$, the relative integration error against $N_\kappa = 28$ is 3.02×10^{-5}. Refining isotropic BEM surface-panel subdivisions across $(n_r, n_\phi, n_z) \in \{(1,1,1),(2,2,2),(3,3,2),(4,4,2),(5,5,3)\}$ in the resolved gap regime yields a fine-level relative BEM discretization uncertainty of 0.0533, while refining the 3D Hockney–Eastwood FFT GEM solver across grids up to $48 \times 48 \times 32$ yields a fine-level vorticity discretization uncertainty of 0.0529 (plotted as conservative $\pm\sigma_{\rm BEM}$ and $\pm\sigma_{\rm vox}$ shaded error bands in Figs. 4 and 7).

IV. PHENOMENON I: SYMMETRY-FILTERED SPIN-2 VERNIER GRAVITATIONAL RADIATION AND PERTURBATIVE $f(R)$ SCALARON WAVES

A. Elimination of Classical Rigid-Body Quadrupole Background and Vernier Selection Rule

Consider a rigid rotor of classical mass density $\rho_{\rm mass}(r,\phi,z)$ invariant under M-fold discrete rotations $\phi \to \phi + 2\pi/M$. Expanding $\rho_{\rm mass} = \sum_{n\in\mathbb{Z}}\rho_{nM}(r,z)e^{inM\phi}$, its trace-free $l = 2, m = \pm 2$ mass quadrupole moments $I^{\rm mass}_{\pm 2} = \int \rho_{\rm mass}(r,\phi,z)r^2 e^{\pm 2i\phi} r\,\mathrm{d}r\,\mathrm{d}\phi\,\mathrm{d}z$ vanish identically whenever $M \ge 3$, because $nM = \pm 2$ admits no integer solution $n \in \mathbb{Z}$ for $M \ge 3$. Consequently, rotating a rigid C_6 vane assembly at constant angular velocity Ω above a stationary C_4 stator produces identically zero classical $l = 2$ mass-quadrupole gravitational radiation ($\dddot{I}^{{\rm mass},C_6}_{ij} = 0$, verified on our 3D mesh to residual 2.72×10^{-16}); classical mass emission is suppressed to the $l = 6$ duodecapole by an additional relativistic factor $(\Omega L/c)^8$.

As proved directly from cyclic group representations of the surface-scattering operator $\mathbf{M}(\tilde{\kappa};\theta)$ in Appendix B, the renormalized Casimir interaction energy density $\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r},\phi,\tilde{z};\theta)$ couples the $M_{\rm up} = 6$ rotor harmonics and $M_{\rm low} = 4$ stator harmonics nonlinearly, generating azimuthal wave numbers $m = pM_{\rm up} - qM_{\rm low}$ ($p, q \in \mathbb{Z}$). At lowest non-trivial order $(p,q) = (\pm 1, \pm 1)$, this yields $m = \pm(6-4) = \pm 2$:

$$\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r},\phi,\tilde{z};\tilde{t}) = \bar{\mathcal{E}}_0(\tilde{r},\tilde{z}) + 2\,\mathrm{Re}\left[\mathcal{E}_2(\tilde{r},\tilde{z})\,e^{i(2\phi - 6\tilde{\Omega}\tilde{t})}\right] + \ldots, \quad (19)$$

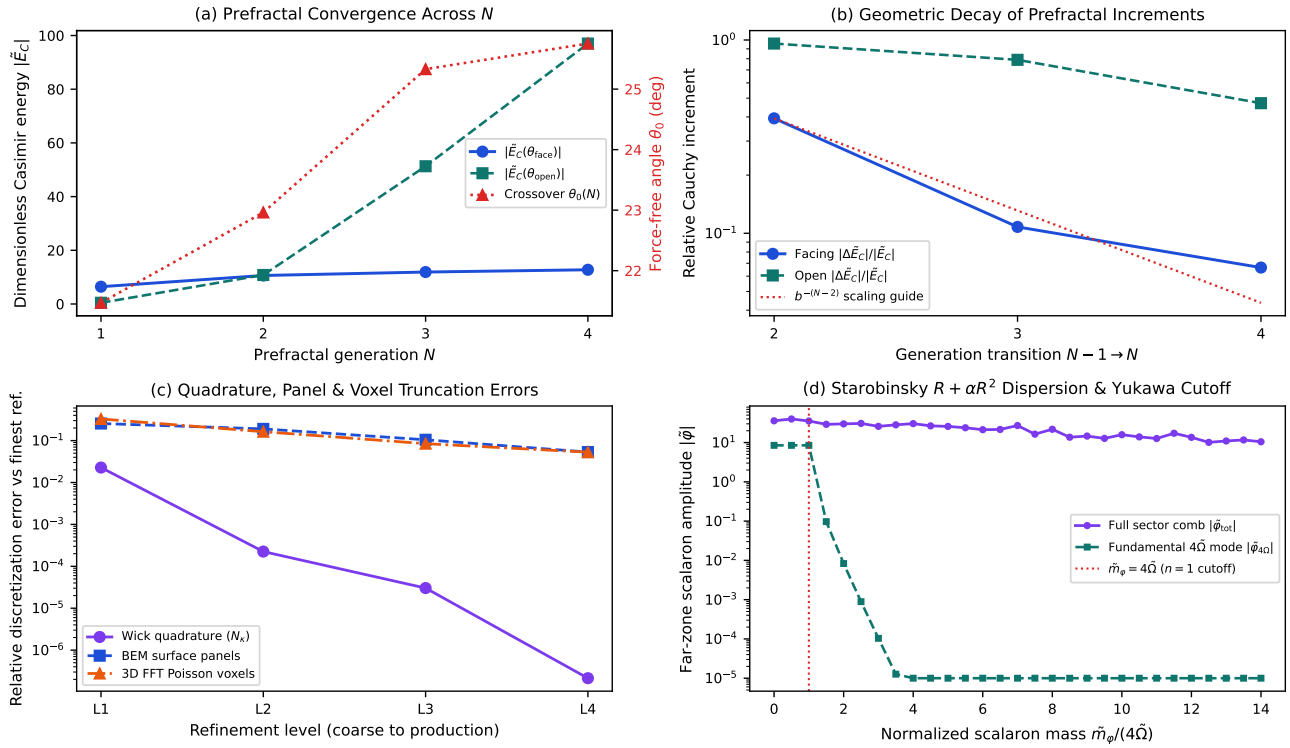


FIG. 2. Numerical verification, prefractal order convergence ($N = 1$–4), discretization error bounds, and massive $f(R)$ scalaron dispersion in dimensionless Planck–Casimir units. (a) Numerical BEM convergence of Casimir interaction energy magnitudes $|\tilde{E}_C(\theta_{\rm face})|$ (solid circles) and $|\tilde{E}_C(\theta_{\rm open})|$ (dashed squares), alongside the force-free crossover angle $\theta_0(N)$ (dotted triangles, right axis), across prefractal generations $N = 1, 2, 3, 4$ ($b = 3$). (b) Semilogarithmic decay of relative Cauchy increments $|\tilde{E}_C(N) - \tilde{E}_C(N-1)|/|\tilde{E}_C(N)|$ versus generation transition $N-1 \to N$ compared against the analytical $b^{-(N-2)}$ geometric scaling guide (dotted red). (c) Monotonic decay of relative truncation errors across refinement levels L1–L4 for Gauss–Legendre Wick quadrature ($N_\kappa \in \{6, 10, 14, 20\}$ vs. $N_\kappa = 28$ reference), isotropic BEM surface-panel subdivisions ((n_r, n_ϕ, n_z) up to $(5, 5, 3)$ in the resolved separation regime), and 3D Hockney–Eastwood FFT Poisson voxel grids ($16\times16\times12$ through $36\times36\times26$ vs. $48 \times 48 \times 32$ reference). (d) Far-zone Starobinsky $f(R) = R + \alpha R^2$ ($\alpha > 0$) monopole scalaron amplitude $|\tilde{\varphi}|$ at $\tilde{r}_{\rm obs} = 10$ versus normalized scalaron mass $\tilde{m}_\varphi/(4\tilde{\Omega})$ comparing the full multi-harmonic sector-edge trace comb $|\tilde{\varphi}_{\rm tot}|$ (solid purple circles) against the fundamental $n = 1$ ($4\tilde{\Omega}$) breathing mode $|\tilde{\varphi}_{4\Omega}|$ (dashed teal squares), demonstrating propagating Klein–Gordon waves below threshold and exponential Yukawa screening above cutoff.

where $\tilde{\Omega} \equiv \Omega L/c \ll 1$. Physically, sectors at $\phi = 0, \pi$ occupy the attractive facing phase ($\Delta\tilde{T}_{00}^{\rm vac} < 0$) while orthogonal sectors at $\phi = \pm\pi/2$ occupy the open-aperture saddle-point phase, sweeping an $m = 2$ quantum-vacuum energy wave at pattern speed $\Omega_{\rm pat} = 3\Omega$ while the angle-integrated energy modulates only at $\mathrm{lcm}(6, 4)\Omega = 12\Omega$ (zero net shaft torque at 6Ω). Crucially, because $6\Omega \ll \pi c/d_0$, the cavity evolves adiabatically in its instantaneous ground state without exciting dynamical Casimir photons; conversely, as shown in Appendix B, placing the stationary $(6, 4)$ cavity in an incoming external GW recovers the parametric resonance structure of de Oliveira *et al.* [35] and GW work coupling of Sorge [34] when $\omega_{\rm GW} = 6\Omega$. Substituting Eq. (19) into Eq. (17) yields circularly polarized on-axis ($\vartheta = 0$) spin-2 waveforms at $\omega_{\rm GW} = 6\Omega$:

$$\tilde{h}_+(\tilde{t}, \tilde{r}, \vartheta) = -\frac{72\,\tilde{\Omega}^2 |\tilde{Q}_2^{(6\Omega)}|}{\tilde{r}} \frac{1 + \cos^2\vartheta}{2} \cos\big(6\tilde{\Omega}(\tilde{t} - \tilde{r}) - \delta_2\big),$$
$$\tilde{h}_\times(\tilde{t}, \tilde{r}, \vartheta) = -\frac{72\,\tilde{\Omega}^2 |\tilde{Q}_2^{(6\Omega)}|}{\tilde{r}} \cos\vartheta \sin\big(6\tilde{\Omega}(\tilde{t} - \tilde{r}) - \delta_2\big), \tag{20}$$

with dimensionless Peters–Mathews power $\tilde{P}_{\rm GW}^{(2,2)} = \frac{46656}{5}\tilde{\Omega}^6 |\tilde{Q}_2^{(6\Omega)}|^2$ (physical luminosity $P_{\rm GW} \propto G\hbar^2\Omega^6 L^2/c^7$), attaining $\tilde{h}_+ = 0.1145$ for PEC EM and 0.1888 for $\xi = 1/6$ at $N = 3$, versus 0.2969 for the single-scale periodic control (ratio 0.3857× due to radial Cantor area partitioning; Fig. 3b).

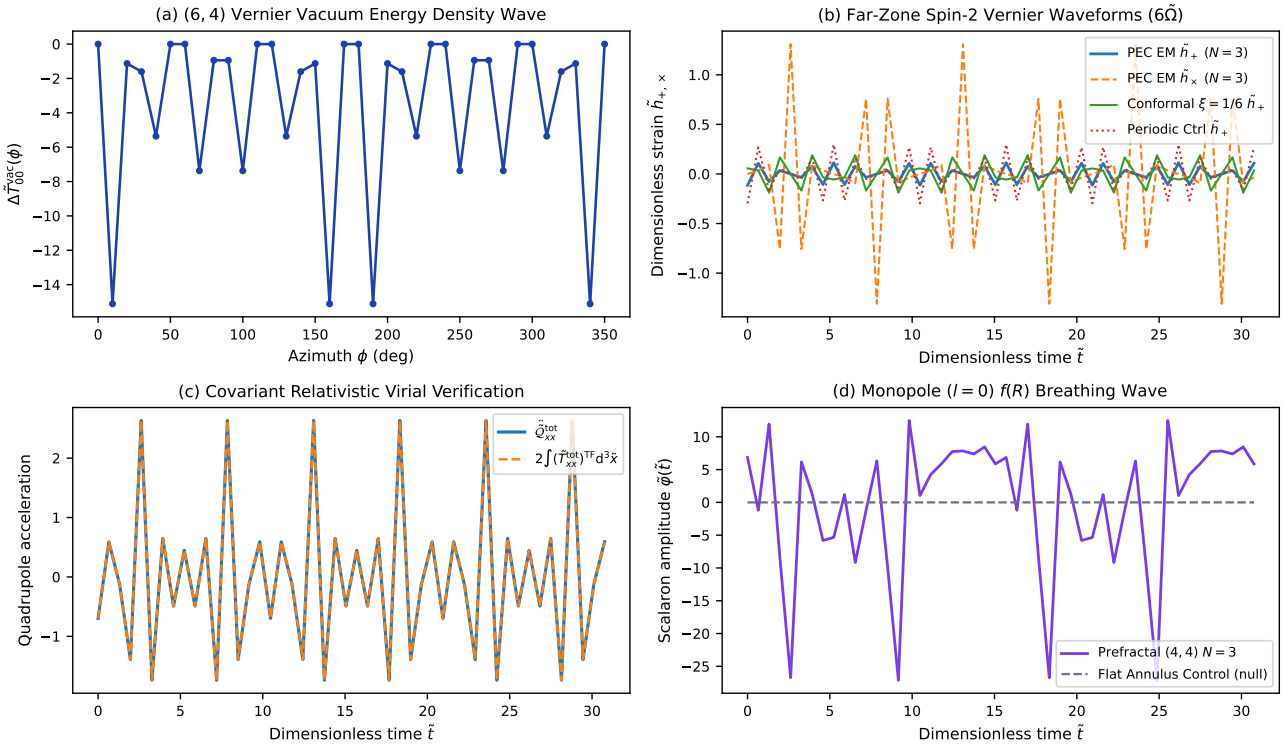


FIG. 3. Symmetry-filtered tensor and scalar-tensor gravitational radiation generated by steady adiabatic rotation ($\tilde{\Omega} = 0.10, \tilde{r}_{\rm obs} = 10$) of the CPSC in dimensionless units ($\epsilon_P = \ell_P/L$). (a) Numerical four-pass BEM reconstruction of the azimuthal Casimir vacuum energy wave $\Delta\tilde{T}^{\rm vac}_{00}(\phi)$ in the $(6,4)$ Vernier cavity ($N = 3$), exhibiting a pure $m = 2$ spatial quadrupole while classical rigid C_6 and C_4 mass quadrupoles vanish to machine precision (2.72×10^{-16}). (b) Numerical far-zone spin-2 waveforms $\tilde{h}_+(\tilde{t})$ and $\tilde{h}_\times(\tilde{t})$ oscillating at $\tilde{\omega}_{\rm GW} = 6\tilde{\Omega}$ for PEC EM (solid blue $\tilde{h}_+$, dashed orange $\tilde{h}_\times$), conformal scalar $\xi = 1/6$ (solid green), and equal-area periodic grating control (dotted red). (c) Numerical verification of the relativistic second-moment virial theorem (A2), comparing $\ddot{\tilde{Q}}^{\rm tot}_{xx}$ (solid) against $2\int(\tilde{T}^{\rm tot}_{xx})^{\rm TF}\mathrm{d}^3\tilde{x}$ (dashed; relative error 1.45×10^{-15}). (d) Monopole ($l = 0$) Starobinsky $f(R)$ breathing waveform $\tilde{\varphi}(\tilde{t})$ in the massless propagation limit for the $(4,4)$ prefractal cavity ($N = 3$, purple) versus the identically silent flat-annulus analytical/numerical control (dashed gray).

B. Perturbative Field Equations and Yukawa Dispersion of the $f(R)$ Scalaron

To formulate the scalar breathing mode rigorously [13, 15–17], consider quadratic Starobinsky gravity $f(R) = R + \alpha R^2$ with $\alpha > 0$ (ensuring both ghost freedom $f'(0) = 1 > 0$ and Dolgov–Kawasaki tachyon stability $f''(0) = 2\alpha > 0$). Varying the metric action yields the exact modified field equations [15, 17]

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} + \big(g_{\mu\nu}\Box - \nabla_\mu\nabla_\nu\big)f'(R) = \frac{8\pi G}{c^4}T^{\rm tot}_{\mu\nu}, \tag{21}$$

whose trace $3\Box f'(R) + f'(R)R - 2f(R) = (8\pi G/c^4)T^{\mu}{}_{\mu,\rm tot}$ linearizes around Minkowski spacetime ($R^{(0)} = 0$) in terms of the canonically normalized scalaron $\varphi \equiv \sqrt{3/2}\,\delta f'(R) = \sqrt{6}\,\alpha\,\delta R$ ($\alpha_\varphi = 1/\sqrt{6}$) with effective mass $m_\varphi = 1/\sqrt{6\alpha} > 0$ ($\tilde{m}_\varphi = L/\sqrt{6\alpha}$) to the Klein–Gordon wave equation

$$\big(\tilde{\Box} - \tilde{m}_\varphi^2\big)\,\tilde{\varphi}(\tilde{t},\tilde{\mathbf{x}}) = 8\pi\alpha_\varphi\,\tilde{T}^{\mu}{}_{\mu,\rm tot}(\tilde{t},\tilde{\mathbf{x}}). \tag{22}$$

By the trace of the virial identity (A2), if an exterior mechanical support frame were infinitely rigid with instantaneous relativistic sound speed, static boundary reaction stresses would cancel $\int_{\rm cav}\tilde{P}^{\rm vac}_i\mathrm{d}^3\tilde{x}$; however, for any physical support ring whose elastic sound speed $c_s \ll c$ places its acoustic relaxation rate $\tau_{\rm el}^{-1} \sim c_s/L$ below the 4Ω breathing modulation of Constellation A $(4,4)$ (where external drive power $\dot{\tilde{E}}_C = -\tilde{\Omega}\tilde{\tau}_\theta(\tilde{t})$ pumps the cavity energy), the uncompensated oscillatory cavity trace $\tilde{\mathcal{T}}_{\rm eff}(\tilde{t}) = \int_{\rm cav}\tilde{T}^{\mu}{}_{\mu,\rm eff}\mathrm{d}^3\tilde{x} = \sum_{n=1}^{\infty}\tilde{\mathcal{T}}_{4n}\cos(4n\tilde{\Omega}\tilde{t})$ sources Eq. (22) via Eq. (18). In the flat-annulus control, $\tilde{T}^{\mu}{}_{\mu,\rm eff} = 0$ (numerical waveform amplitude 0). In the $(4,4)$ prefractal CPSC ($N = 3$), $\tilde{\varphi}$ attains amplitude 27.122 in the light-scalaron limit ($\tilde{m}_\varphi \ll 4\tilde{\Omega}$; Fig. 3d), whereas for a heavy Starobinsky scalaron ($\tilde{m}_\varphi = 2\times4\tilde{\Omega}$, small $\alpha > 0$) the fundamental $4\tilde{\Omega}$ mode becomes evanescent and decays at $\tilde{r}_{\rm obs} = 10$ to $9.80\times10^{-4}\times$ its massless amplitude, while sharp sector-edge harmonics $4n\tilde{\Omega}$ peel off sequentially across the angular-transit comb before total Yukawa screening sets in (Fig. 2d).

V. PHENOMENON II: FORCE-FREE SPACETIME CURVATURE CROSSOVER AND TOLMAN–KOMAR ACTIVE MASS

In a stationary asymptotically flat spacetime possessing timelike Killing vector $\xi^\mu = (1, \mathbf{0})$ and unit hypersurface normal n^μ, the general-relativistic Komar active gravitational mass enclosed within radius $r_{\rm enc}$ reduces at linear order ($\sqrt{-g} \simeq 1$, $g_{00} \simeq -1$) in $(-,+,+,+)$ signature to Tolman's active mass formula [2, 3, 31, 54, 55]:

$$\begin{aligned}\Delta\tilde{M}_{\rm Komar}(\tilde{r}_{\rm enc};\theta) &= 2\int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}} \left(\tilde{T}^{\rm tot}_{\mu\nu} - \frac{1}{2}\eta_{\mu\nu}\tilde{T}^{\alpha}{}_{\alpha,\rm tot}\right) n^\mu \xi^\nu \mathrm{d}^3\tilde{x} \\ &= \int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}} \left(2\tilde{T}^{\rm tot}_{00} + \tilde{T}^{\mu}{}_{\mu,\rm tot}\right) \mathrm{d}^3\tilde{x} \\ &= \int_{|\tilde{\mathbf{x}}|<\tilde{r}_{\rm enc}} \Big(\tilde{\rho}_{\rm tot} + \sum_{i=x,y,z} \tilde{P}_{i,\rm tot}\Big) \mathrm{d}^3\tilde{x}. \end{aligned} \tag{23}$$

For a complete, static, self-equilibrated cavity-plus-support system ($\partial_j \tilde{T}^{ij}_{\rm tot} = 0$ with support ring inside $\tilde{r}_{\rm sup}$), the identity $\tilde{T}^{ij}_{\rm tot} = \partial_k(\tilde{x}^i \tilde{T}^{kj}_{\rm tot})$ guarantees $\int_{\mathbb{R}^3} \tilde{P}_{i,\rm tot}\, \mathrm{d}^3\tilde{x} = 0$, recovering $\Delta\tilde{M}_{\rm Komar}(\infty;\theta) = \tilde{E}_C(\theta)$ in exact agreement with Fulling *et al.* [28] and Sorge [31]. Within the prefractal cavity, however, two distinct gravitational features emerge:

1. *Force-free curvature crossover ($\theta = \theta_0$)*: Between open-aperture repulsion $\tilde{F}_z(0,\tilde{z}) > 0$ [50] and facing-sector attraction $\tilde{F}_z(\theta_{\rm face},\tilde{z}) < 0$, continuity guarantees an angle $\theta_0(\tilde{z})$ where net axial force vanishes:

$$\tilde{F}_z(\theta_0,\tilde{z}) = -\frac{\partial\tilde{E}_C}{\partial|\tilde{z}|}\bigg|_{\theta_0} = \frac{\tilde{A}_{\rm ann}}{|\tilde{z}|^4}\Big[3\mathcal{C}(\theta_0) - \partial_{\ln|\tilde{z}|}\mathcal{C}(\theta_0)\Big] = 0. \tag{24}$$

   At $\theta_0 = 25.3298°$ (force residual $|\tilde{F}_z| = 0.0504$), the axial support constraint vanishes completely while $\tilde{E}_C(\theta_0) < 0$ remains finite, locking $\partial_{\ln|\tilde{z}|}\mathcal{C}|_{\theta_0} = 3\mathcal{C}(\theta_0)$ (numerical ratio 3.0018). Substituting into Eq. (11) yields a finite *force-free coarse-grained Ricci curvature*:

$$\tilde{R}_{\rm eff}(\theta_0,\tilde{z}) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta_0,\tilde{z}) = \frac{8\pi}{|\tilde{z}|^4}\Big[3\mathcal{C}(\theta_0) + \partial_{\ln\tilde{R}}\mathcal{C}(\theta_0)\Big], \tag{25}$$

   attaining $\tilde{R}_{\rm eff}(\theta_0) = -2317.8935$ at $N = 3$ (Fig. 4b,c).

2. *Quasi-local interior Tolman pressure integral versus asymptotic WEP mass*: When the mechanical support ring sits at exterior radius $\tilde{r}_{\rm sup} = 1.25 > \tilde{r}_{\rm out}$, a quasi-local integration surface $\tilde{r}_{\rm enc} \in [\tilde{r}_{\rm in}, \tilde{r}_{\rm out}]$ encloses the uncompensated interior vacuum pressure $\int_{\rm cav}\sum_i \tilde{P}_{i,\rm vac}\mathrm{d}^3\tilde{x} \neq 0$, whereas integrating past $\tilde{r}_{\rm sup}$ cancels spatial stresses and yields a pure WEP binding-mass shift $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C(\theta_{\rm face}) - \tilde{E}_C(0) = 39.4077$ matching the near-zone equilibrated swing 39.4065 to 3×10^{-5} at fixed baryon number (Fig. 4d).

VI. PHENOMENON III: DISCRETE-SCALE-INVARIANT LOG-PERIODIC SPACETIME METRIC COMB

In critical gravitational collapse, Choptuik [47] showed that discrete self-similarity produces log-periodic echoing in spacetime curvature corresponding to complex scaling exponents [46, 48]. In our homothetically staged CPSC (6), static Lorenz-gauge Einstein theory gives $\tilde{\nabla}^2\tilde{h}_{00}(\tilde{\mathbf{x}}) = -8\pi\big(\tilde{\rho}_{\rm tot} + \sum_i \tilde{P}_{i,\rm tot}\big) = -8\pi\big(2\tilde{\rho}_{\rm tot} + \tilde{T}^{\mu}{}_{\mu,\rm tot}\big)$. Along the collision-free homothetic open-aperture slot-center ray $\tilde{\mathbf{x}}(\tilde{z}) = \big(\tilde{r}_{\rm in} + \frac{1}{2}\Delta\tilde{r}(|\tilde{z}|/\tilde{Z}_1), 0, -1.044|\tilde{z}|\big)$ with upper vane elevation $\tilde{z}_{\rm vane} = -0.956|\tilde{z}|$, rescaling $|\tilde{z}| \to |\tilde{z}|/b$ maps the scattered Green's function of generation m onto generation $m+1$ across $\tilde{\ell}_* \ll |\tilde{z}| \ll \tilde{Z}_1$. Mellin transformation of the Newtonian Green's integral $\tilde{h}_{00}(\tilde{\mathbf{x}};\tilde{z}) = 2\int|\tilde{\mathbf{x}} - \tilde{\mathbf{x}}'|^{-1}(2\tilde{\rho}_{\rm tot} + \tilde{T}^{\mu}{}_{\mu,\rm tot})\mathrm{d}^3\tilde{x}'$ yields complex scaling exponents $\Delta_k = 2 + i\omega_k$ ($\omega_k = 2\pi k/\ln b$):

$$\tilde{h}_{00}(\tilde{\mathbf{x}};\theta_{\rm open},\tilde{z}) = \frac{1}{|\tilde{z}|^2}\sum_{k=-\infty}^{\infty}\mathcal{H}_k^{(00)}\left(\frac{\tilde{\mathbf{x}}}{|\tilde{z}|}\right)\exp\left(i\frac{2\pi k}{\ln b}\ln\frac{|\tilde{z}|}{\tilde{\ell}_*}\right), \tag{26}$$

while the axial tidal Riemann component $\tilde{R}_{0z0z} = -\frac{1}{2}\partial_{\tilde{z}}^2\tilde{h}_{00}$ scales as $|\tilde{z}|^{-4}$ with quadratically amplified harmonic weights $(2+i\omega_k)(3+i\omega_k)\mathcal{H}_k^{(00)}$. Rotating to $\theta_{\rm face}$ engages facing-sector confinement and quenches the sidewall comb. Across $N = 4$ homothetic Cantor stages ($b = 3$), harmonic regression on $\delta\ln(|\tilde{z}|^2|\tilde{h}_{00}|)$ recovers $\hat{\omega}_1 = 5.7096$ and $\hat{b} = 3.0055$ (0.17% and 0.18% from exact $2\pi/\ln 3 = 5.7192$ and $b = 3$, respectively; $R^2 = 0.6301$ vs. 0.2411 in the periodic control, peak open tidal curvature 1585.3059, and rotary harmonic suppression 3.1177×; Fig. 5).

VII. PHENOMENON IV: ROTARY ENERGY-CONDITION SWITCHING, RAYCHAUDHURI GEODESIC LENSING, AND WORLDLINE QEIS

At linear order in ϵ_P^2, test null congruences ($k^\mu = (1,\hat{\mathbf{k}})$) and slow timelike congruences ($u^\mu = (1,\mathbf{0})$) with zero initial twist and shear obey the Raychaudhuri equa-

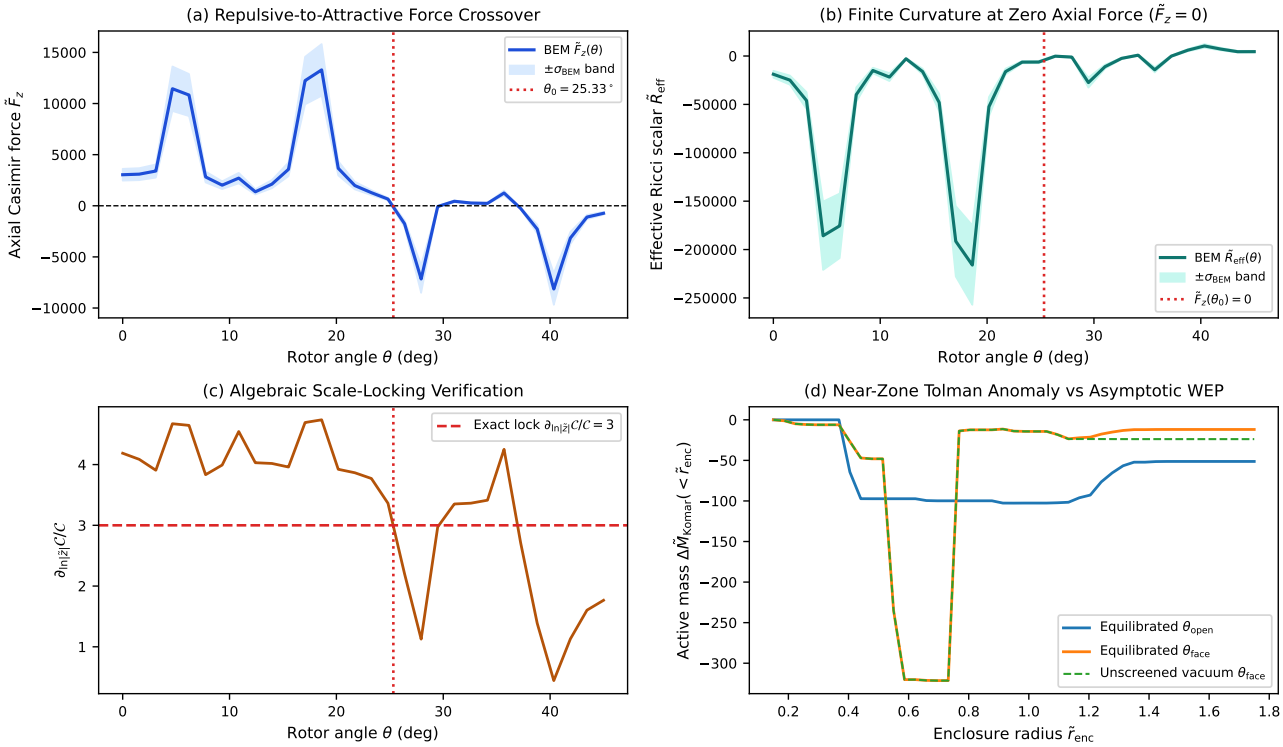


FIG. 4. Decoupling of axial mechanical boundary force from coarse-grained spacetime curvature and radial resolution of the Tolman–Komar active gravitational mass in Constellation A $(4,4)$ ($N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical BEM sweep of dimensionless axial Casimir force $\tilde{F}_z(\theta)$ with $\pm\sigma_{\rm BEM}$ discretization uncertainty band (shaded blue), crossing continuously from open-aperture repulsion ($\tilde{F}_z > 0$) to facing-sector attraction ($\tilde{F}_z < 0$) at $\theta_0 = 25.3298^\circ$ (vertical dotted line). (b) Numerical coarse-grained Ricci curvature $\tilde{R}_{\rm eff}(\theta) = -8\pi\tilde{T}^{\mu}{}_{\mu,\rm eff}(\theta)$ with $\pm\sigma_{\rm BEM}$ error band (shaded teal), remaining non-zero (-2317.8935) at $\tilde{F}_z(\theta_0) = 0$. (c) Numerical verification of the analytical scale-locking condition $\partial_{\ln|\tilde{z}|}\mathcal{C}/\mathcal{C} = 3$ (dashed red horizontal line) at $\theta = \theta_0$. (d) Cumulative radial integral of the active gravitational mass $\Delta\tilde{M}_{\rm Komar}(<\tilde{r}_{\rm enc})$ from Eq. (23), contrasting unscreened interior vacuum stresses (dashed green) against self-equilibrated cavities supported at $\tilde{r}_{\rm sup} = 1.25$ in open (blue) and facing (orange) states, recovering $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C(\theta)$ asymptotically.

tions [2, 18, 19]:

$$\frac{\mathrm{d}\tilde{\Theta}_{\rm null}}{\mathrm{d}\tilde{\lambda}} = -\tilde{R}_{\mu\nu}k^\mu k^\nu = -8\pi\underbrace{\left(\tilde{T}_{00} + \tilde{T}_{ij}\hat{k}^i\hat{k}^j\right)}_{\widetilde{\rm NEC}_{\hat{\mathbf{k}}}}, \tag{27}$$

$$\frac{\mathrm{d}\tilde{\Theta}_{\rm time}}{\mathrm{d}\tilde{\tau}} = -\tilde{R}_{\mu\nu}u^\mu u^\nu = -4\pi\underbrace{\left(2\tilde{T}_{00} + \tilde{T}^{\mu}{}_{\mu}\right)}_{\widetilde{\rm SEC}}. \tag{28}$$

Whereas null focusing is blind to $\tilde{T}^{\mu}{}_{\mu}$ ($g_{\mu\nu}k^\mu k^\nu = 0$), timelike focusing couples directly to $\tilde{T}^{\mu}{}_{\mu}$, distinguishing $\xi = 1/6$ from $\xi = 0$ via $-\frac{1}{2}\tilde{\Box}\langle\hat{\phi}^2\rangle_{\rm ren}$:

1. *Rotary switching between defocusing and focusing*: At $\theta_{\rm face}$, mode exclusion gives $\tilde{T}_{00} < 0$ and $\tilde{T}_{zz} \approx 3\tilde{T}_{00} < 0$, so $\widetilde{\rm NEC}_z$ reaches -579.895 and axial rays defocus ($\mathrm{d}\tilde{\Theta}_{\rm null}/\mathrm{d}\tilde{\lambda} > 0$). At $\theta_{\rm open}$, transverse slot fringing ($\langle E_r^2\rangle > \langle E_z^2\rangle$) renders $\tilde{T}_{zz} = \tilde{P}_z > |\tilde{\rho}_{\rm vac}| > 0$, flipping $\widetilde{\rm NEC}_z$ up to 6523.7421 and $\widetilde{\rm SEC} \geq 0$ across volume fraction 0.9959, yielding net null expansion swing 100.8036 and trace-to-net timelike focusing ratio 40.2844 at $\xi = 0$ (Fig. 6a–c).

2. *Strict compliance with Ford–Roman and Fewster worldline QEIs*: As emphasized by Fewster [40, 41] and Ford and Roman [38, 39], local negative Casimir densities cannot violate quantum energy inequalities along timelike observer worldlines. A stationary gap observer at $(\tilde{r}_0, \phi_0, \tilde{z}_0)$ (wall clearance $\tilde{d}_{\rm wall} = \tilde{d}_0/2$) samples $\tilde{T}_{00}(\tilde{t}) = \tilde{T}_{00}(\tilde{r}_0, \phi_0 - \tilde{\Omega}\tilde{t}, \tilde{z}_0)$ over sub-reflection Lorentzian timescale $\tilde{\tau} = (\tilde{\tau}_\Omega^{-2} + \tilde{d}_{\rm wall}^{-2})^{-1/2}$ ($\tilde{\tau}_\Omega = \Delta\phi_{\rm vane}/(M_{\rm up}\tilde{\Omega})$), obeying [38, 39, 41]

$$\tilde{\mathcal{I}}_{\rm QEI}(\tilde{\tau}) \equiv \int_{-\infty}^{\infty} \tilde{T}_{00}(\tilde{t})\frac{2\tilde{\tau}^3/\pi}{(\tilde{t}^2+\tilde{\tau}^2)^2}\,\mathrm{d}\tilde{t} \geq -\frac{C_{\rm FR}}{\tilde{\tau}^4}, \tag{29}$$

with $C_{\rm FR} = 3/(32\pi^2)$ per massless scalar ($C_{\rm FR}^{\rm EM} = 3/(16\pi^2)$ for EM; note the normalized unit-area Lorentzian weight $\int \frac{2\tilde{\tau}^3/\pi}{(\tilde{t}^2+\tilde{\tau}^2)^2}\mathrm{d}\tilde{t} = 1$). Across two decades of $\tilde{\Omega}$, the maximum QEI saturation ratio is

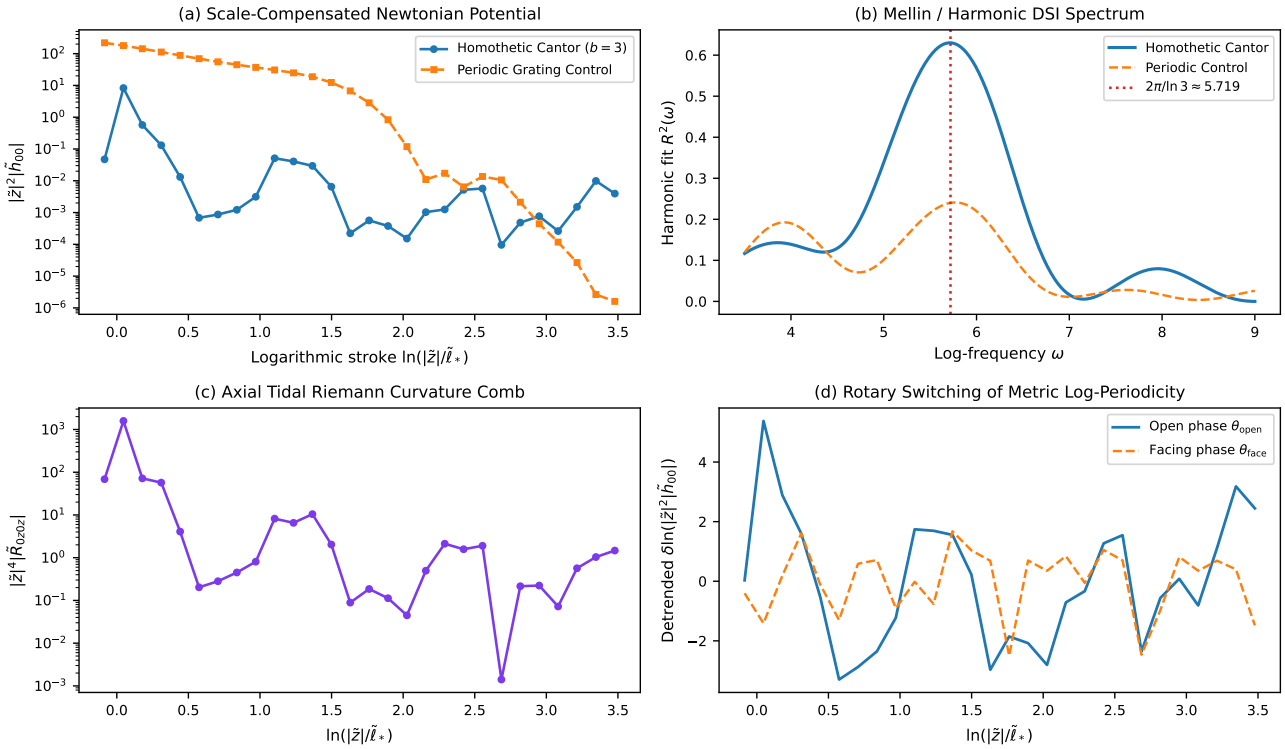


FIG. 5. Imprinting discrete scale invariance onto linearized spacetime geometry via the homothetically staged prefractal CPSC ($b = 3, N = 4$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical BEM–Poisson evaluation of the scale-compensated Newtonian metric modulus $|\tilde{z}|^2|\tilde{h}_{00}(\tilde{\mathbf{x}}(\tilde{z});\tilde{z})|$ swept logarithmically across $\ln(|\tilde{z}|/\tilde{\ell}_*)$ along the collision-free homothetic slot-center ray for the Cantor hierarchy (blue circles) versus the equal-area periodic grating control (orange squares). (b) Harmonic regression spectrum $R^2(\omega)$ of the affine-detrended log-modulus $\delta\ln(|\tilde{z}|^2|\tilde{h}_{00}|)$ for the homothetic Cantor cavity (solid blue, peaking at $\hat{\omega}_1 = 5.7096$) and periodic control (dashed orange) against the exact theoretical DSI frequency $\omega_1 = 2\pi/\ln 3 \approx 5.719$ (vertical dotted red line). (c) Numerical scale-compensated axial tidal Riemann curvature modulus $|\tilde{z}|^4|\tilde{R}_{0z0z}|$ across Cantor stages. (d) Rotary suppression of detrended log-periodic metric oscillations upon rotating from $\theta_{\rm open}$ (solid blue) to $\theta_{\rm face}$ (dashed orange).

$0.1425 < 1$ (Fig. 6d), ruling out unphysical exotic-matter instabilities.

VIII. PHENOMENON V: CANTOR-STRATIFIED GRAVITOMAGNETIC FRAME DRAGGING FROM BROKEN BOOST INVARIANCE

As clarified in the gravito-electromagnetic (GEM) reviews of Mashhoon [23] and Costa and Natário [24, 25], the term "frame dragging" must never be conflated with viscous fluid dragging or trivial coordinate transformations to a rotating frame. Instead, working strictly in the *non-rotating asymptotic Minkowski laboratory frame* (t, x, y, z), Costa and Natário [25] classify physical frame dragging into three distinct gravitomagnetic levels: (i) the gravitomagnetic 1-form vector potential $\tilde{h}_{0i}$ (governing Sagnac clock-synchronization shifts), (ii) the coordinate-gauge-invariant gravitomagnetic field $\tilde{\mathbf{B}}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0$ (governing physical Coriolis-like Lense–Thirring precession $\tilde{\boldsymbol{\Omega}}_{\rm LT} = -\frac{1}{2}\tilde{\mathbf{B}}_g$ of a stationary Fermi–Walker-transported gyroscope relative to distant stars), and (iii) the gravitomagnetic tidal tensor $\tilde{\mathbb{H}}_{ij} = -\frac{1}{2}\partial_{(i}\tilde{B}_{g,j)}$. In linearized Lorenz gauge, $\tilde{h}_{0i}$ is sourced by laboratory momentum density $\tilde{T}^{0i} = -\tilde{T}_{0i}$:

$$\tilde{\nabla}^2\tilde{h}_{0i}(\tilde{\mathbf{x}}) = -16\pi\tilde{T}_{0i}(\tilde{\mathbf{x}}) = +16\pi\tilde{T}^{0i}(\tilde{\mathbf{x}}), \qquad \tilde{\mathbf{B}}_g = \tilde{\nabla}\times\tilde{\mathbf{h}}_0. \tag{30}$$

Why does a rotating flat-disk Casimir cavity carry zero vacuum angular momentum, whereas a rotating CPSC actively drags stationary gyroscopes? Under a local tangential Lorentz boost $\tilde{v}_\phi = \tilde{\Omega}\tilde{r} \ll 1$ from the boundary rest frame to the inertial laboratory frame, a rest-frame stress tensor transforms to laboratory momentum density

$$\tilde{T}^{0\phi}_{\rm vac}(\tilde{\mathbf{x}}) = \frac{\tilde{v}_\phi}{1-\tilde{v}_\phi^2}\big(\tilde{\rho}_{\rm vac} + \tilde{P}_\phi\big) \simeq \tilde{\Omega}\tilde{r}\big(\tilde{T}_{00} + \tilde{T}_{\phi\phi}\big) + \mathcal{O}(\tilde{\Omega}^3), \tag{31}$$

corresponding to net axial vacuum angular momentum $\tilde{J}_z^{\rm vac} = \tilde{\Omega}\int\tilde{r}^2(\tilde{T}_{00}+\tilde{T}_{\phi\phi})\mathrm{d}^3\tilde{x}$. For planar plates (2), boost invariance enforces $\tilde{T}_{00} + \tilde{T}_{\phi\phi} = \tilde{\rho}_{\rm vac} + \tilde{P}_\parallel = 0$ identically (numerical enthalpy residual 0), so $\tilde{T}^{0\phi}_{\rm vac} = 0$ and $\tilde{J}_z^{\rm vac} = 0$. In Constellation C, radial and chiral sector walls break ϕ-boost symmetry, producing a sign-alternating azimuthal

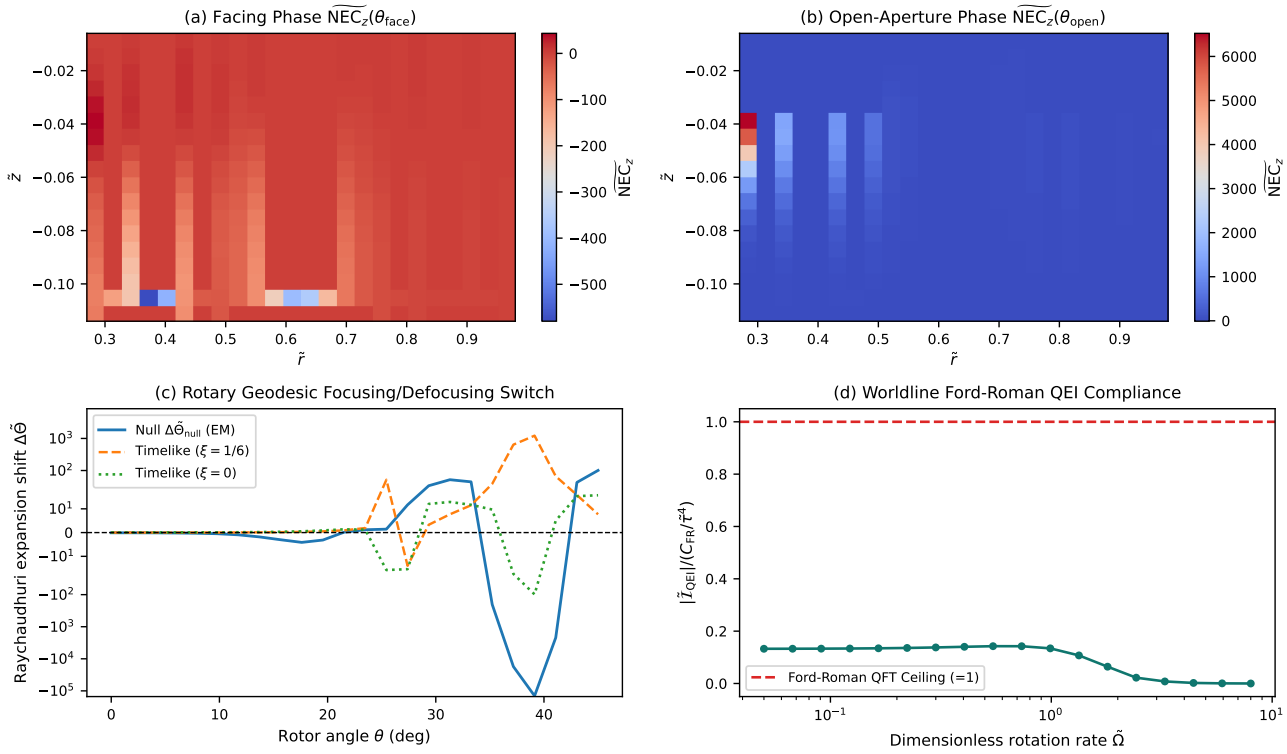


FIG. 6. Rotary switching of general-relativistic energy conditions, Raychaudhuri geodesic focusing, and worldline Quantum Energy Inequality (QEI) compliance in Constellation A $(4, 4)$ ($N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a,b) Numerical four-pass BEM voxel maps in the $(\tilde{r}, \tilde{z})$ meridian plane of the axial Null Energy Condition scalar $\widetilde{\mathrm{NEC}}_z = \tilde{T}_{00} + \tilde{T}_{zz}$ in (a) the defocusing facing-sector phase θ_{face} ($\widetilde{\mathrm{NEC}}_z < 0$) and (b) the converging open-aperture phase θ_{open} ($\widetilde{\mathrm{NEC}}_z > 0$). (c) Numerical Raychaudhuri geodesic expansion shifts $\Delta\tilde{\Theta}$ (symmetric logarithmic scale) across $\theta \in [0°, 45°]$ comparing null rays in PEC EM (solid blue) against timelike trajectories for conformal ($\xi = 1/6$, dashed orange) and minimal ($\xi = 0$, dotted green) scalar fields. (d) Numerical Lorentzian-sampled worldline QEI ratio $|\tilde{I}_{\mathrm{QEI}}|/(C_{\mathrm{FR}}/\tilde{\tau}^4)$ versus rotation rate $\tilde{\Omega}$ (solid teal circles) bounded strictly below the analytical Ford–Roman QFT ceiling (= 1, dashed red line).

vacuum enthalpy $\tilde{w}_\phi \equiv \tilde{T}_{00} + \tilde{T}_{\phi\phi} \neq 0$ across the 2^{m-1} Cantor slot tracks. Solving Eq. (30) in the inertial laboratory frame—thereby solving the active counterpart to Zhang's stationary test-field problem [36]—yields a Cantor-stratified Lense–Thirring vortex lattice with 4 radial vorticity sign reversals, peak normalized gyroscope precession $|\tilde{\Omega}^{\mathrm{vac}}_{\mathrm{LT},z}|/\tilde{\Omega} = 22931.2037$ at $N = 3$, and a $23.0732\times$ enhancement in root-mean-square gravitomagnetic vorticity over the matched periodic grating (Fig. 7).

IX. DISCUSSION: COVARIANT CONSISTENCY, EQUIVALENCE PRINCIPLE, AND PLANCK-SCALE MAGNITUDE

A. Role of Constraint Stresses and Elimination of Coordinate Artifacts

Evaluating linearized Einstein or Komar integrals over $\langle\hat{T}_{\mu\nu}\rangle_{\mathrm{vac}}$ without mechanical support stresses $\tilde{T}^{\mathrm{con}}_{\mu\nu}$ violates $\partial_\mu \tilde{T}^{\mu\nu}_{\mathrm{tot}} = 0$ and introduces gauge artifacts [27, 28, 31]. Enforcing Eq. (13) explicitly across all five phenomena clarifies how boundary reactions enter each sector:

1. In the $(6, 4)$ Vernier radiator (Sec. IV), elastic support stresses balance local pressures; however, because $\tilde{\mathcal{Q}}^{\mathrm{tot}}_{ij}$ integrates energy density $\tilde{T}_{00}$—where rigid C_6 and C_4 mass distributions have identically zero $m = 2$ quadrupole and elastic strain energy scales as $\tilde{P}^2/Y_{\mathrm{rigid}} \to 0$ in the rigid limit—the 6Ω spin-2 wave is genuinely sourced by the sweeping $m = 2$ Casimir vacuum energy wave, verified to virial error 1.45×10^{-15}.

2. In the stationary Tolman–Komar sector (Sec. V), exterior support stresses enforce $\int_{\mathbb{R}^3} \tilde{P}_{i,\mathrm{tot}} \mathrm{d}^3\tilde{x} = 0$ globally so $\Delta\tilde{M}_{\mathrm{Komar}}(\infty) = \tilde{E}_C(\theta)$ [28, 31], while leaving interior coarse-grained curvature $\tilde{R}_{\mathrm{eff}}(\theta_0)$ at zero axial force unscreened inside the open gap.

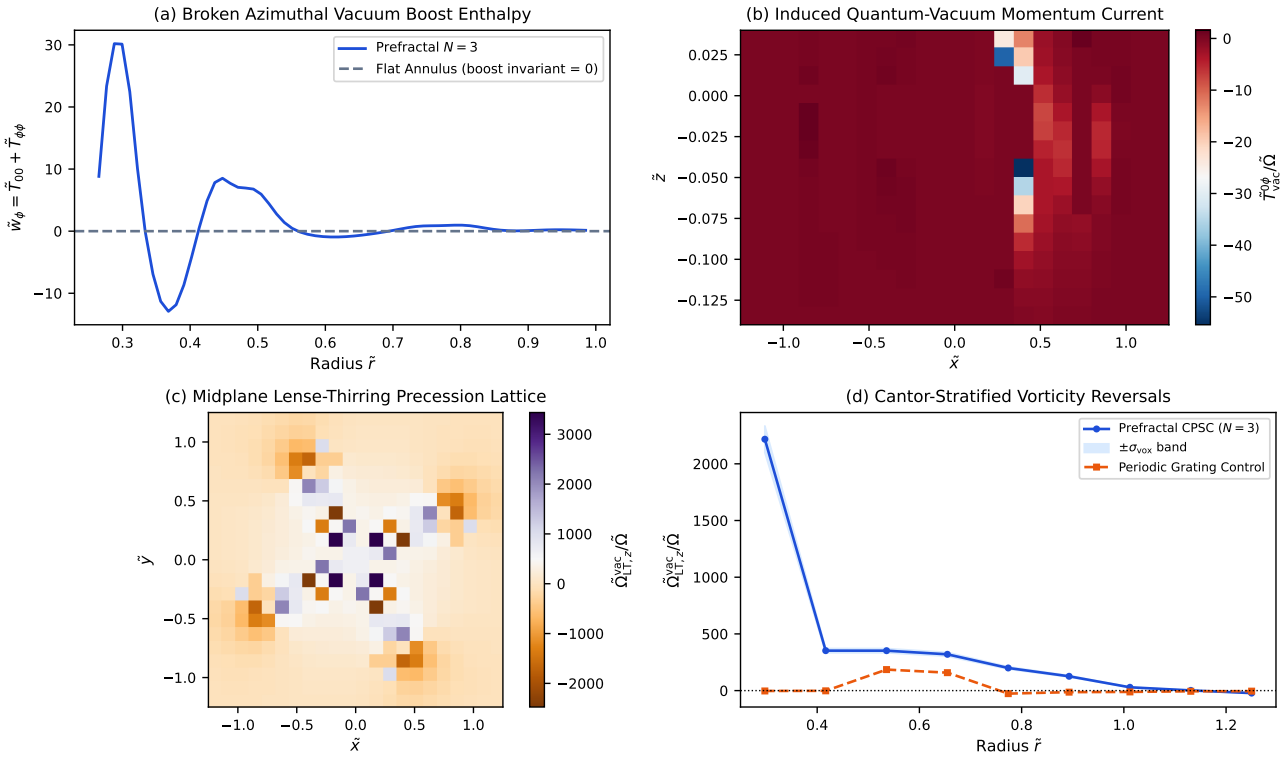


FIG. 7. Gauge-invariant gravitomagnetic Lense–Thirring precession generated by broken azimuthal Lorentz boost invariance in Constellation C (chiral $(4,4)$, $N = 3$, dimensionless units $\epsilon_P = \ell_P/L$). (a) Numerical radial profile of the rest-frame azimuthal vacuum enthalpy $\tilde{w}_\phi = \tilde{T}_{00} + \tilde{T}_{\phi\phi}$ for the prefractal CPSC (solid blue) versus the boost-invariant flat-annulus control ($\tilde{w}_\phi = 0$, dashed gray). (b) Cross-sectional $(\tilde{x}, \tilde{z})$ map of the laboratory-frame azimuthal vacuum momentum current $\tilde{T}^{0\phi}_{\rm vac}/\tilde{\Omega}$. (c) Midplane $(\tilde{x}, \tilde{y})$ lattice of the dimensionless Lense–Thirring gyroscope precession rate $\tilde{\Omega}^{\rm vac}_{{\rm LT},z}/\tilde{\Omega} = -\frac{1}{2}\tilde{B}^{\rm vac}_{g,z}/\tilde{\Omega}$ obtained from the 3D spectral GEM Poisson solver (30). (d) Radial slice of $\tilde{\Omega}^{\rm vac}_{{\rm LT},z}/\tilde{\Omega}$ with $\pm\sigma_{\rm vox}$ discretization band (shaded blue) showing 4 Cantor vorticity sign reversals compared against the matched periodic grating control (dashed orange squares).

TABLE I. Summary of the five active semiclassical gravitational phenomena of the Canonical Prefractal Sector Cavity (CPSC) in Planck-normalized dimensionless units ($\epsilon_P = \ell_P/L$), contrasting analytical/numerical flat-annulus limits, equal-area single-scale periodic controls, and prefractal Cantor hierarchies ($b = 3$).

Gravitational Phenomenon	Governing Coupling	Flat Annulus	Periodic Control	Prefractal CPSC
I(a). Spin-2 Vernier GWs $(6,4)$	$\ddot{\tilde{\mathcal{Q}}}^{\rm tot}_{ij}$ ($\tilde{h}_+$ amp.)	0 (axisym.)	0.2969	0.1145 (0.3857×)
I(b). Spin-0 Scalaron Wave $(4,4)$	$\tilde{\mathcal{T}}_{\rm eff}(t)$ $(R + \alpha R^2, \alpha > 0)$	0	Single-band	27.122 (Yukawa 9.80×10^{-4}×)
II. Force-Free Curvature ($\tilde{F}_z = 0$)	$\tilde{R}_{\rm eff}(\theta_0)$, $\theta_0 = 25.3298°$	Forbidden ($\tilde{F}_z < 0$)	Single crossover	−2317.8935 (lock 3.0018)
III. Log-Periodic Metric Comb	$\hat{\omega}_1$ (R^2 at $2\pi/\ln b$)	Power-law (0)	$R^2 = 0.2411$	$\hat{\omega}_1 = 5.7096$ ($R^2 = 0.6301$)
IV. NEC/SEC & Geodesic Switch	$\widetilde{\rm NEC}_z$ / QEI sat.	Violated ($\forall\tilde{\mathbf{x}}$)	Single switch	$[-579.895, 6523.7421]$; QEI 0.1425
V. Vacuum Frame Dragging	$\lvert\tilde{\Omega}^{\rm vac}_{{\rm LT},z}\rvert/\tilde{\Omega}$	0 (boost inv.)	Ref. vortex	22931.2037 (23.0732×, 4 rev.)

B. Physical Scale Estimates and Analog-Gravity Connections

Restoring physical constants at an illustrative outer radius $L = 1$ μm ($\epsilon_P^2 = \ell_P^2/L^2 = 2.612 \times 10^{-58}$) underscores the purely foundational nature of these effects in real spacetime: static metric perturbations scale as $|h_{\mu\nu}| = \epsilon_P^2|\tilde{h}_{\mu\nu}| \sim 10^{-56}$–$10^{-54}$, coarse-grained Ricci curvatures scale as $|R_{\rm eff}| = (\epsilon_P^2/L^2)|\tilde{R}_{\rm eff}| \sim 10^{-43}$ m^{-2}, and Vernier strains scale as $h_+ \sim \epsilon_P^2(L/r_{\rm obs})(\Omega L/c)^2|\tilde{Q}_2|$, far below direct gravitational-wave detection thresholds even for synchronized $N_{\rm arr} \times N_{\rm arr}$ arrays. Nevertheless, in acoustic Bose–Einstein condensate and optical metamaterial analog-gravity platforms where effective propagation speeds $c_{\rm eff} \ll c$ and effective couplings replace ℓ_P, analogous boundary-induced effective-trace and boost-breaking vorticity phenomena offer promising targets for laboratory simulation.

X. CONCLUSION

We have formulated a self-contained semiclassical field-theoretic and numerical framework for prefractal Casimir gravitation at zero temperature, uniting three-dimensional boundary-element scattering with linearized Einstein, Starobinsky $f(R)$, and gravito-electromagnetic theory. Strictly separating the cavity-integrated macroscopic effective trace [12] from local curved-spacetime conformal trace anomalies [7, 8] and enforcing covariant stress-energy conservation [28, 31], we demonstrated five active gravitational backreaction phenomena in the Canonical Prefractal Sector Cavity (Table I): (1) symmetry-filtered 6Ω Vernier spin-2 gravitational waves and Yukawa-screened $l = 0$ Starobinsky scalaron waves; (2) an axial-force-free macroscopic curvature crossover respecting global Tolman–Komar WEP mass recovery; (3) a switchable discrete-scale-invariant log-periodic Newtonian potential and tidal Riemann comb; (4) rotary NEC/SEC geodesic focusing/defocusing switching strictly bounded by Ford–Roman and Fewster worldline QEIs; and (5) Cantor-stratified Lense–Thirring frame dragging driven by broken azimuthal boost invariance.

Several caveats and future directions merit emphasis. First, our calculations assume idealized zero-temperature ($T = 0$ K) lossless Dirichlet and PEC boundaries in the adiabatic slow-rotation regime ($\Omega L/c \ll 1$); incorporating finite-conductivity Lifshitz dielectric dispersion, nonzero temperature ($T > 0$ K, where the fractal thermal trace $\rho_{\rm th}(3/d_s - 1)$ enters [12, 43]), and non-adiabatic dynamical Casimir particle creation at $\Omega L/c \sim \mathcal{O}(1)$ [35] represents a natural theoretical extension. Second, while real-spacetime amplitudes are suppressed by $\epsilon_P^2 \sim 10^{-58}$ at micrometer length scales, mapping the Vernier selection rules and broken-boost enthalpy currents onto metamaterial and superfluid analog-gravity systems provides a concrete avenue for future experimental exploration.

DATA AVAILABILITY

All simulation scripts, three-dimensional boundary-element and spectral linearized gravity solvers, provenance ledgers, figure sidecars, and raw JSON datasets supporting the findings of this study are included in the standalone `fractal_casimir_gravity` software package and are available from the author upon reasonable request.

ACKNOWLEDGMENTS

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Appendix A: Covariant Virial Identity and Conservation of the Total Quadrupole Tensor

In linearized general relativity around Minkowski spacetime, Lorenz gauge $\partial^\mu \bar{h}_{\mu\nu} = 0$ requires $\partial_\mu T^{\mu\nu}_{\rm tot} = 0$:

$$\frac{\partial \tilde{T}^{00}_{\rm tot}}{\partial \tilde{t}} + \partial_j \tilde{T}^{j0}_{\rm tot} = 0, \qquad \frac{\partial \tilde{T}^{0i}_{\rm tot}}{\partial \tilde{t}} + \partial_j \tilde{T}^{ji}_{\rm tot} = 0. \tag{A1}$$

Multiplying the energy continuity equation by $\tilde{x}^i \tilde{x}^j$, integrating over a ball $\mathcal{B}_R$ enclosing the isolated cavity-plus-support assembly ($\tilde{T}^{\mu\nu}_{\rm tot} = 0$ on $\partial\mathcal{B}_R$), and integrating by parts twice using momentum continuity yields the relativistic second-moment virial theorem:

$$\frac{\mathrm{d}^2}{\mathrm{d}\tilde{t}^2} \int_{\mathcal{B}_R} \tilde{T}^{00}_{\rm tot}(\tilde{\mathbf{x}}, \tilde{t})\, \tilde{x}^i \tilde{x}^j \, \mathrm{d}^3\tilde{x} = 2 \int_{\mathcal{B}_R} \tilde{T}^{ij}_{\rm tot}(\tilde{\mathbf{x}}, \tilde{t})\, \mathrm{d}^3\tilde{x}. \tag{A2}$$

In the ideal rigid-support limit ($Y_{\rm rigid} \to \infty$), elastic displacements scale as $\delta\tilde{u} \sim \tilde{P}/Y_{\rm rigid} \to 0$, so elastic strain energy vanishes ($\tilde{T}^{00}_{\rm con,elas} \sim \tilde{P}^2/Y_{\rm rigid} \to 0$) while constraint stresses $\tilde{T}^{ij}_{\rm con} \sim \mathcal{O}(\tilde{P})$ satisfy $\partial_j \tilde{T}^{ji}_{\rm con} = -\partial_j \tilde{T}^{ji}_{\rm vac} - \partial_{\tilde{t}} \tilde{T}^{0i}_{\rm tot}$ at boundary walls. For static configurations ($\partial_{\tilde{t}} = 0$), Eq. (A2) gives $\int \tilde{T}^{ij}_{\rm con} \mathrm{d}^3\tilde{x} = -\int \tilde{T}^{ij}_{\rm vac} \mathrm{d}^3\tilde{x}$ (Laue's theorem), ensuring $\Delta\tilde{M}_{\rm Komar}(\infty) = \tilde{E}_C$ [28, 31]. During steady rotation $\theta(\tilde{t}) = \tilde{\Omega}\tilde{t}$ of the $(6, 4)$ Vernier cavity, $\tilde{T}^{00}_{\rm vac}$ carries the $m = 2$ wave (19) whereas rigid C_6 and C_4 mass densities carry zero $m = 2$ harmonic; by Eq. (A2), $\int (\tilde{T}^{ij}_{\rm vac} + \tilde{T}^{ij}_{\rm con})^{\rm TF} \mathrm{d}^3\tilde{x} = \frac{1}{2} \ddot{\tilde{\mathcal{Q}}}^{\rm vac}_{ij}$, verified numerically to relative error 1.45×10^{-15}.

Appendix B: Scattering-Operator Derivation of the Vernier Multipole Selection Rule and GW Reciprocity

Let $\mathcal{U}(\alpha)$ denote the spatial rotation operator about the z axis by angle α. For an upper boundary $\partial\Omega_{\rm U}(\theta)$ with discrete $C_{M_{\rm up}}$ symmetry rotated by θ and a stationary lower boundary $\partial\Omega_{\rm L}$ with $C_{M_{\rm low}}$ symmetry, the single-body Euclidean scattering matrices satisfy $\mathcal{U}(2\pi/M_{\rm low})\mathbf{\Gamma}_{\rm LL}\mathcal{U}^{-1}(2\pi/M_{\rm low}) = \mathbf{\Gamma}_{\rm LL}$ and $\mathbf{\Gamma}_{\rm UU}(\theta) = \mathcal{U}(\theta)\mathbf{\Gamma}_{\rm UU}(0)\mathcal{U}^{-1}(\theta)$ with $\mathcal{U}(2\pi/M_{\rm up})\mathbf{\Gamma}_{\rm UU}(0)\mathcal{U}^{-1}(2\pi/M_{\rm up}) = \mathbf{\Gamma}_{\rm UU}(0)$. Expanding the off-diagonal propagators in cylindrical azimuthal harmonics $e^{i\ell\phi}$ ($\ell \in \mathbb{Z}$), $\mathbf{\Gamma}^{-1}_{\rm UU}(\theta)$ couples $\ell \to \ell + pM_{\rm up}$ with phase $e^{-ipM_{\rm up}\theta}$ ($p \in \mathbb{Z}$), whereas $\mathbf{\Gamma}^{-1}_{\rm LL}$ couples $\ell' \to \ell' - qM_{\rm low}$ ($q \in \mathbb{Z}$). Inserting this Neumann/Born expansion of $\mathbf{\Gamma}^{-1}(\tilde{\kappa}; \theta)$ into the coincidence Green's correlator (16) shows that every term in the renormalized vacuum energy density takes the exact double-Bloch series form

$$\Delta\tilde{T}^{\rm vac}_{00}(\tilde{r}, \phi, \tilde{z}; \theta) = \sum_{p=-\infty}^{\infty} \sum_{q=-\infty}^{\infty} \mathcal{A}_{p,q}(\tilde{r}, \tilde{z})\, e^{i(pM_{\rm up} - qM_{\rm low})\phi - ipM_{\rm up}\theta}. \tag{B1}$$

Projecting onto radiative spin-s spherical/cylindrical multipoles $e^{\mp is\phi}$ ($s = 2$ for spin-2 quadrupole radiation) enforces the exact selection rule

$$pM_{\rm up} - qM_{\rm low} = \pm s, \qquad \omega_{\rm rad} = |p|M_{\rm up}\Omega. \tag{B2}$$

For $(M_{\rm up}, M_{\rm low}) = (6, 4)$ and $s = 2$, the minimal non-zero integers $(p, q) = (\pm 1, \pm 1)$ satisfy $6(\pm 1) - 4(\pm 1) = \pm 2$ and radiate at $\omega_{\rm GW} = 6\Omega$, whereas any non-interacting rigid body ($q = 0$ or $p = 0$) requires $6p = \pm 2$ or $-4q = \pm 2$, which has no integer solution. Conversely, if the $(6, 4)$ cavity rotates inside an incoming circularly polarized plane gravitational wave $h_+^{\rm ext}(t) = h_0 \cos(\omega_{\rm GW} t)$, the linear interaction Hamiltonian $H_{\rm int}(t) = -\frac{1}{2}\int h_{ij}^{\rm ext}(t) T_{\rm tot}^{ij}(\mathbf{x}, t) {\rm d}^3x = -\frac{1}{4}\ddot{h}_{ij}^{\rm ext}(t)\mathcal{Q}_{ij}^{\rm vac}(t)$ yields a secular phase-locked energy transfer (and permanent memory shift in the sense of Sorge [34]) selectively when $\omega_{\rm GW} = 6\Omega$, while reducing in the high-frequency non-adiabatic limit ($\omega_{\rm GW} \to 2\omega_{\rm cav}$) to the parametric dynamical Casimir resonance derived by de Oliveira *et al.* [35].

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